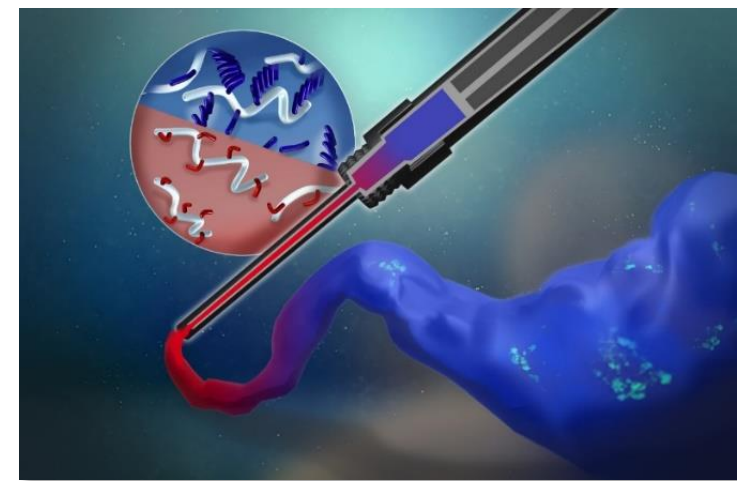
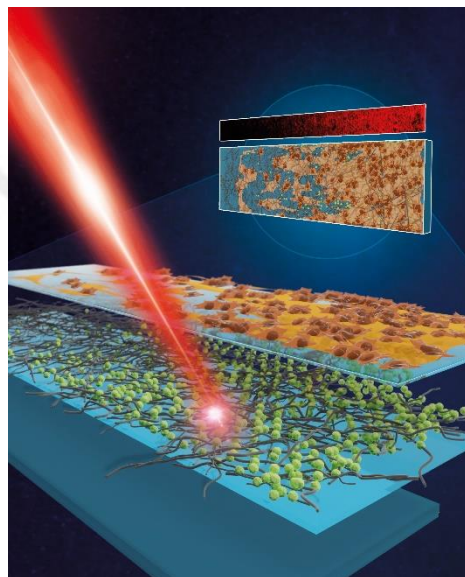
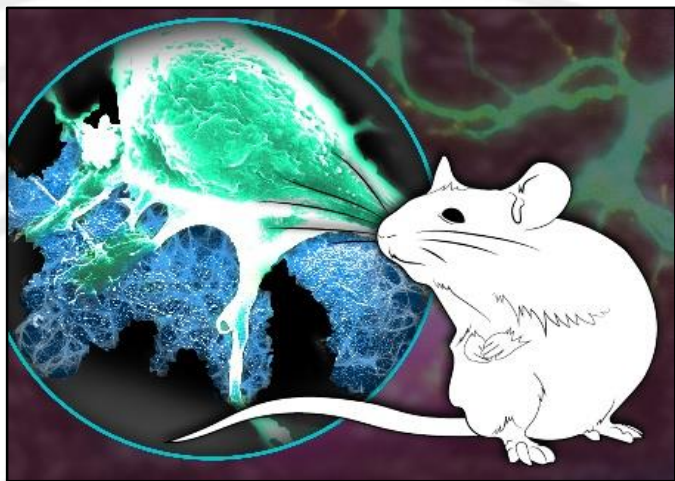


Controlling Cell-Material Interactions through Responsive Peptide Nanostructures



C.V. Synatschke

Max Planck Institute for Polymer Research, Mainz

synatschke@mpip-mainz.mpg.de

Mainz / Rhine Main Area

Historical City



Research and Development



- 30 km from Frankfurt Airport
- Recreational area
- Famous wine region
- High Tech Companies (Hoechst, Clariant, Boehringer, BioNTech, ...)



founded 1984

ca. 550 Employees

6 Departments

6 Departments



Prof. Dr. Paul Blom

Molecular Electronics



Prof. Dr. Mischa Bonn

Molecular
Spectroscopy



**Prof. Dr. Hans-Jürgen
Butt**

Physics of Interfaces



Prof. Dr. Kurt Kremer

Theory of Polymers



**Prof. Dr. Katharina
Landfester**

Physical Chemistry of
Polymers



Prof. Dr. Tanja Weil

Synthesis of
Macromolecules



Prof. Dr. Tanja Weil

Director MPI-P

Department Synthesis of Macromolecules



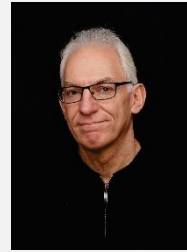
Dr. Manfred Wagner

NMR Spectroscopy



Dr. Markus Klapper

Polymer Synthesis
and Characterization



Dr. Joachim Räder

Mass Spectrometry



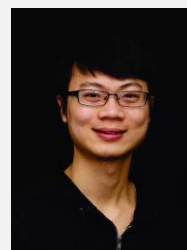
Dr. Seah Ling Kuan

Protein Bioconjugates
and Therapeutics



Dr. David Ng

Life-Like Synthetic
Nanosystems



Dr. Yingke Wu

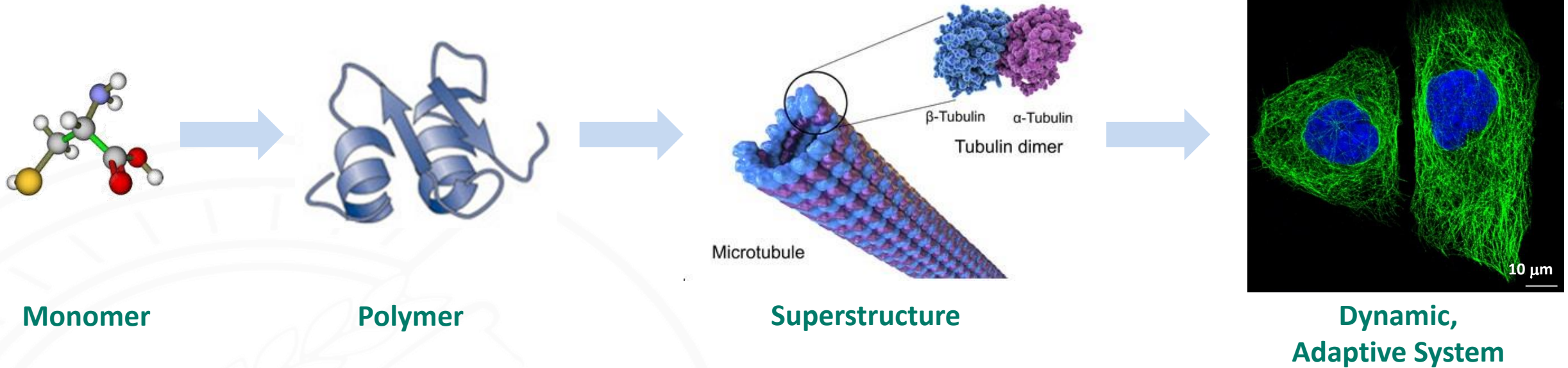
Nanodiamonds as
Biosensors



**Dr. Christopher
Synatschke**

Functional Biomaterials

**„Synthesis of functional macromolecules to tackle future challenges in biomedicine
and material science “**



Material Design in Nature

- Adaptable building blocks
- Internal order through directed interactions
- Multi-scale, hierarchical structuring
- Spatio-temporal control over formation/destruction

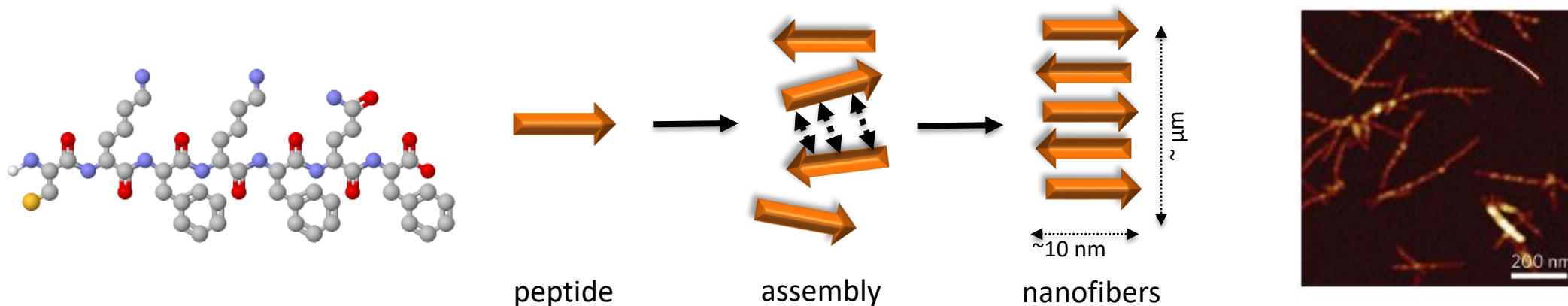
Time as additional Dimension:

Towards “4D Materials”

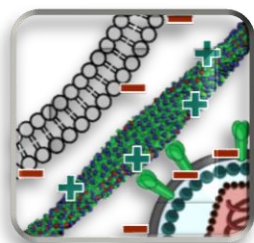
Functional Supramolecular Peptide Nanomaterials



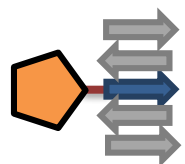
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SAPs for
Modification of
Cell-Virus
Interaction



Data Mining

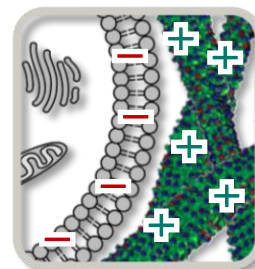


Functionalization



Coassembly

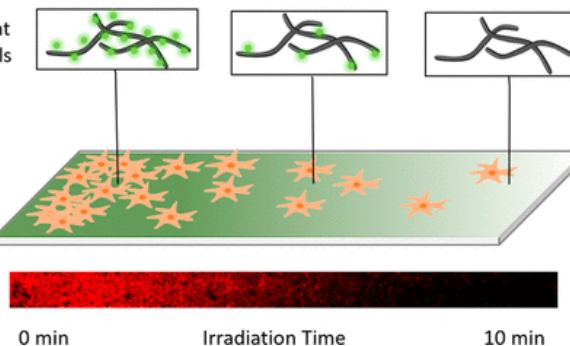
SAPs for
Controlled Cell
Interaction



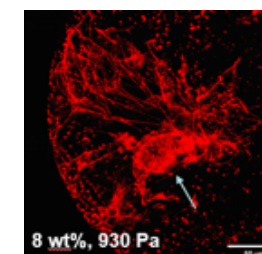
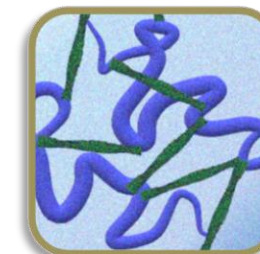
Epitope Gradient
on Peptide Fibrils

Cell Density
Gradient

MALDI-MSI



SAPs as
Gelator for
3D Materials



As ECM for
cell culture



3D Printing

Relevance of Gene Delivery

Introduction of foreign genes into cells

- Research - Biochemistry and Biology
- Pharmaceutical Industry - Biotechnology
- Personalized Medicine - Gene Therapy

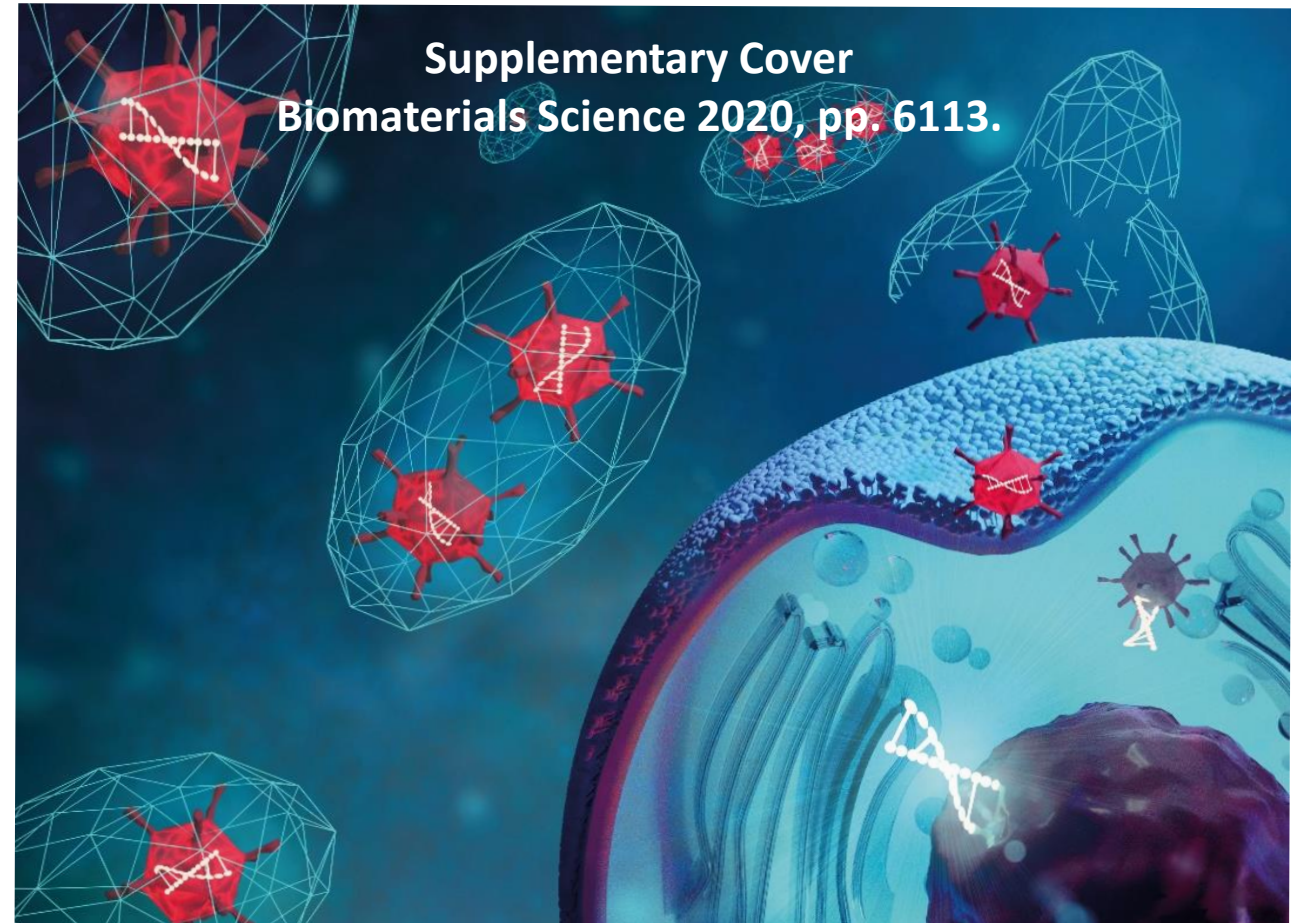
Viruses are Nature's Gene Delivery System

Current Viral Vectors Lack Efficiency

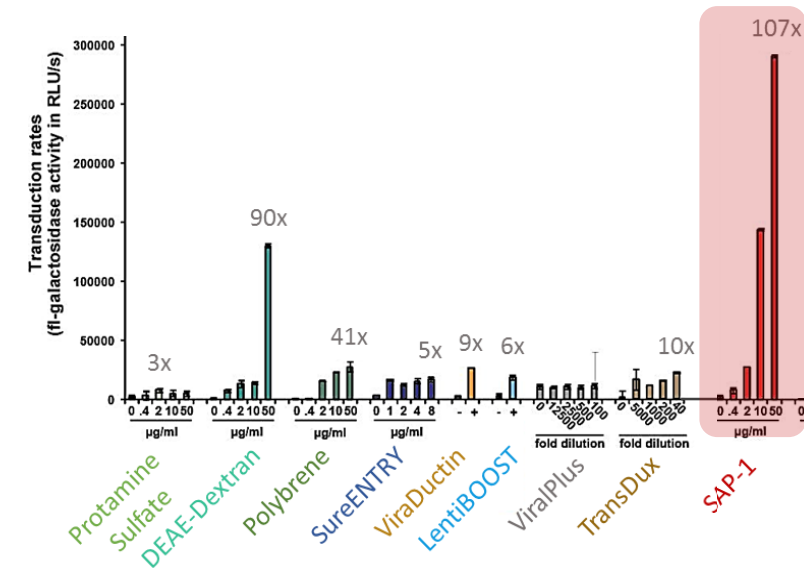
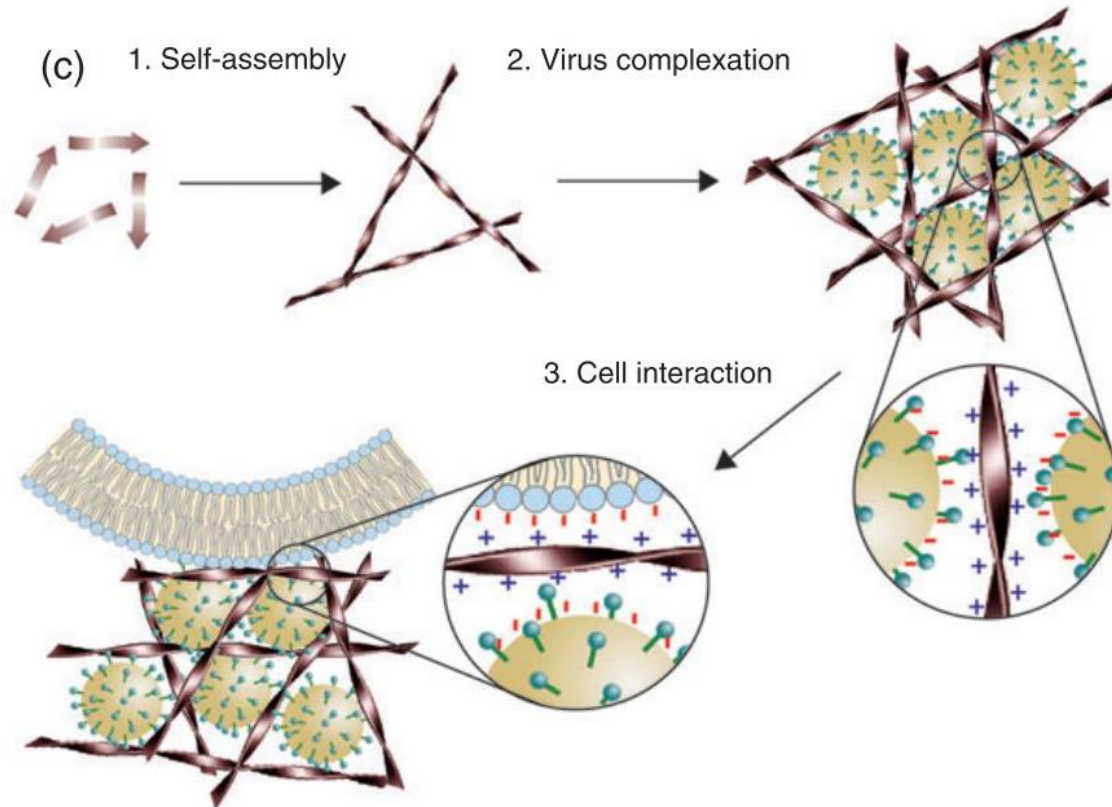
→ Need for transduction enhancing additives

Review Article:

“Materials promoting viral gene delivery”



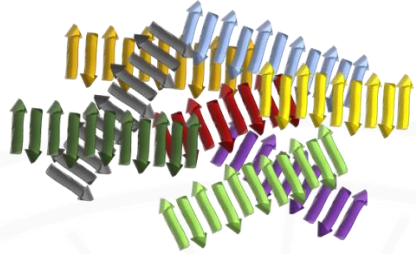
Viral Infectivity Enhancing Peptide Nanofibrils



Jan Münch

Viral Infectivity Enhancing Peptide Nanofibrils

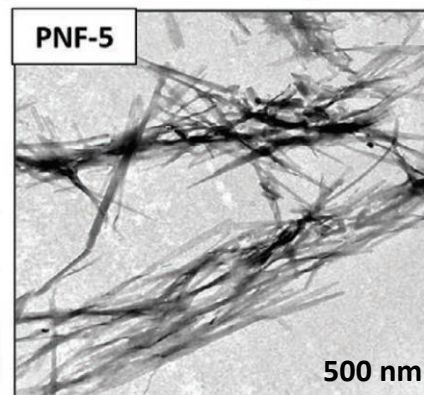
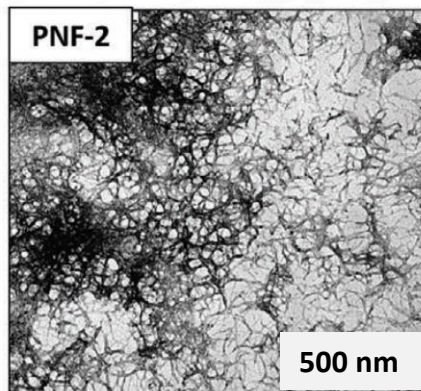
Synthesis + Characterization of SAP library



Requirements:

- Ability to form fibers
- Positive ζ -potential

$\zeta = +19 \text{ mV}$
high activity



$\zeta = +16 \text{ mV}$
low activity



K. Kaygisiz



S. Sieste

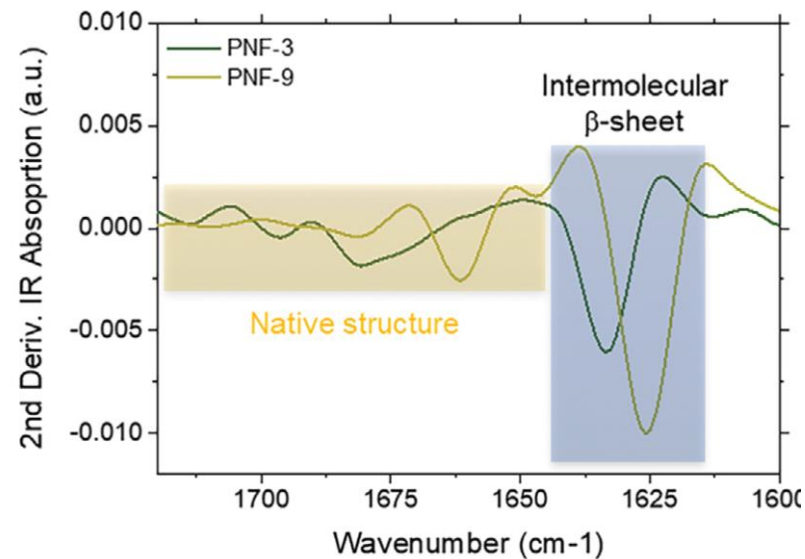
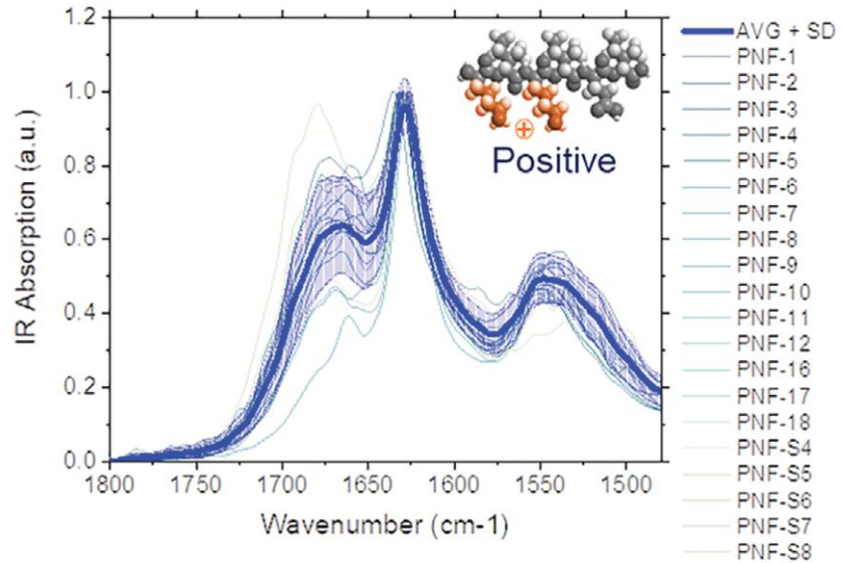


T. Mack

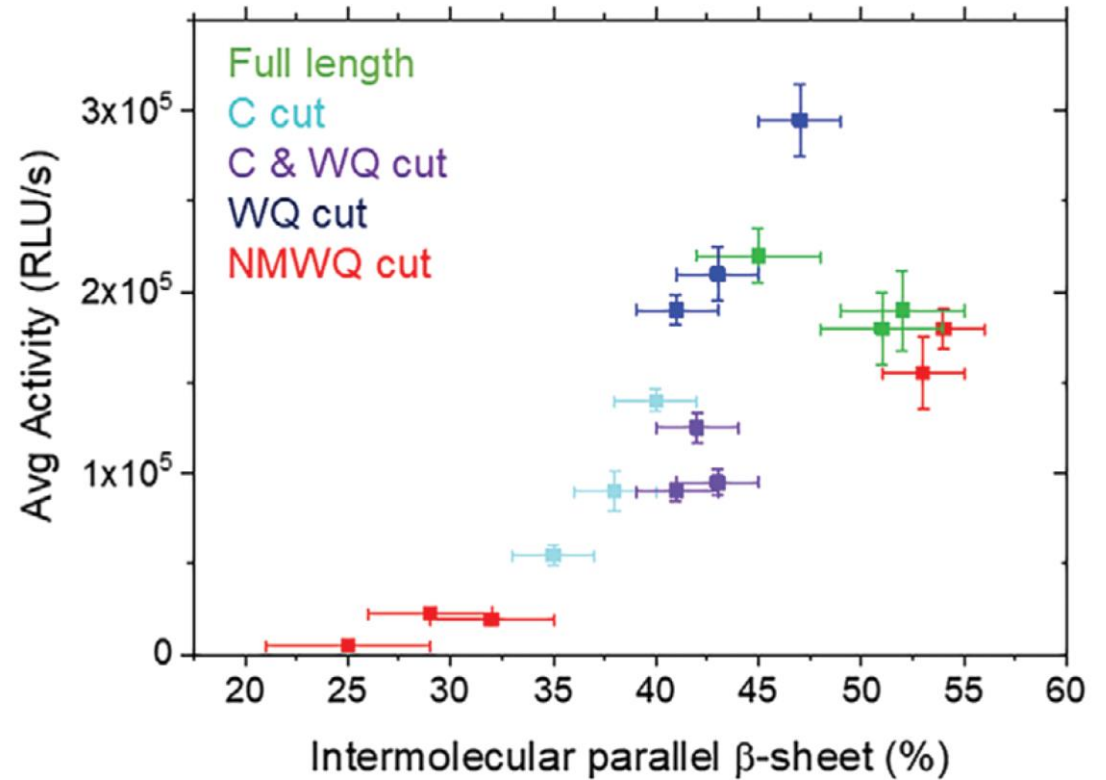


Jan Münch

Internal Structure: Beta Sheet Content



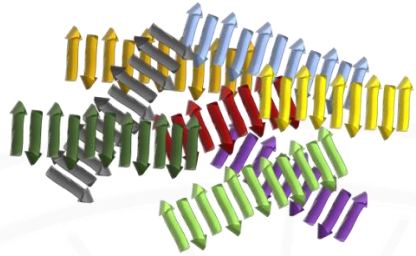
PNF-3: CKIKQII
 PNF-9: CKIKIQI



Peptides with higher beta sheet content tend
 to have higher transduction efficiencies.

Viral Infectivity Enhancing Peptide Nanofibrils

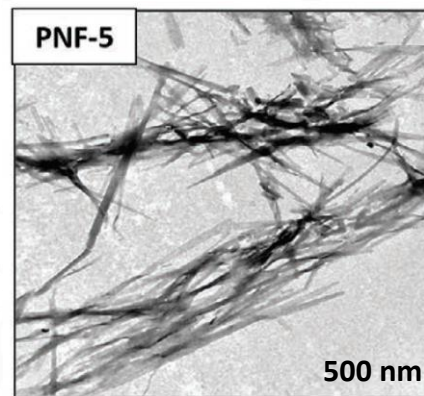
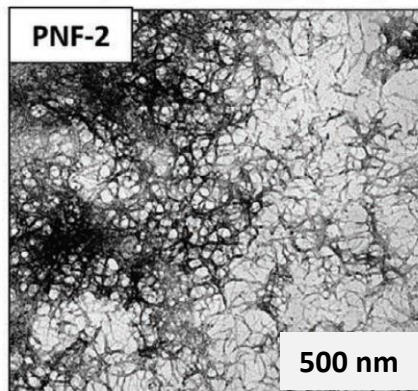
Synthesis + Characterization of SAP library



Requirements:

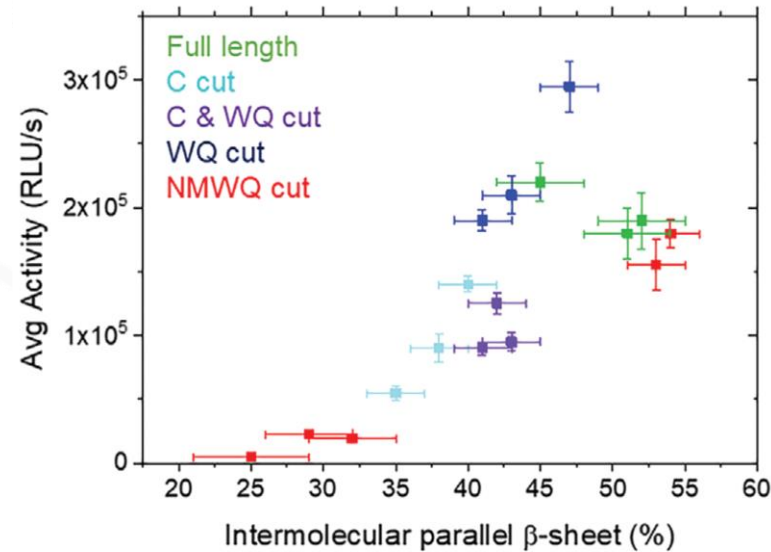
- Ability to form fibers
- Positive ζ -potential

$\zeta = +19$ mV
high activity

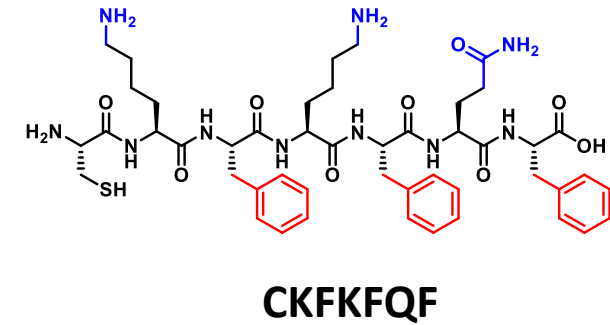


$\zeta = +16$ mV
low activity

Internal Structure



Optimized sequence 45% shorter peptide



K. Kaygisiz



S. Sieste



T. Mack

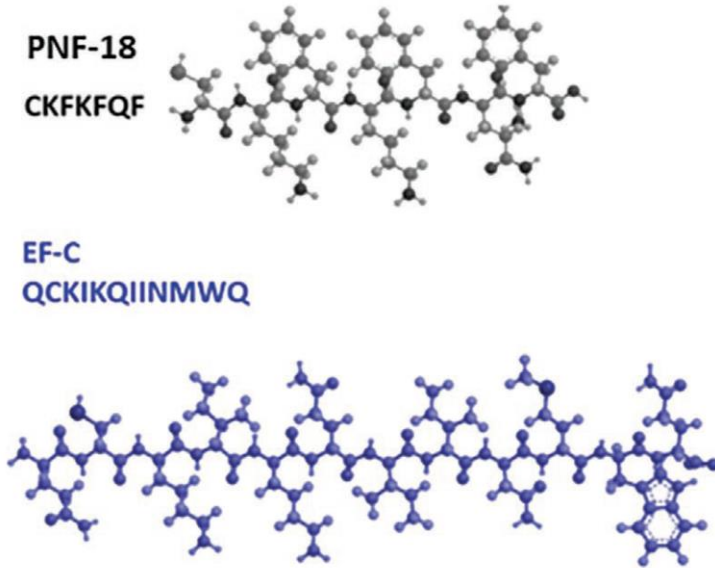


Jan Münch

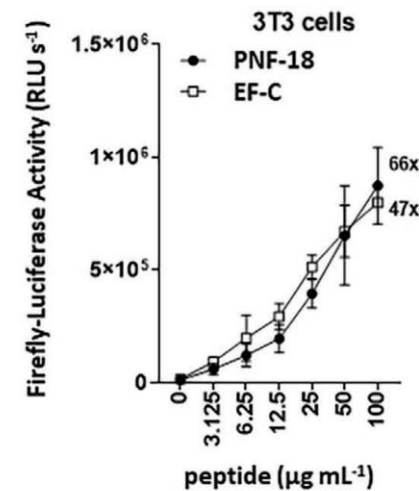
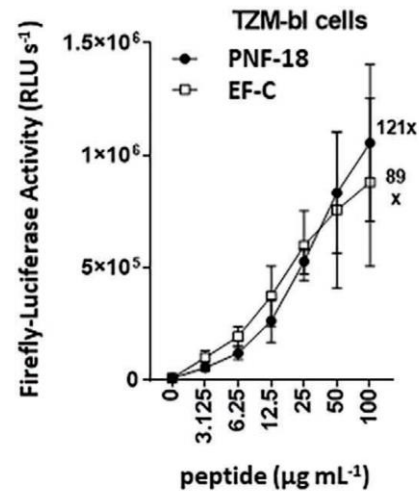
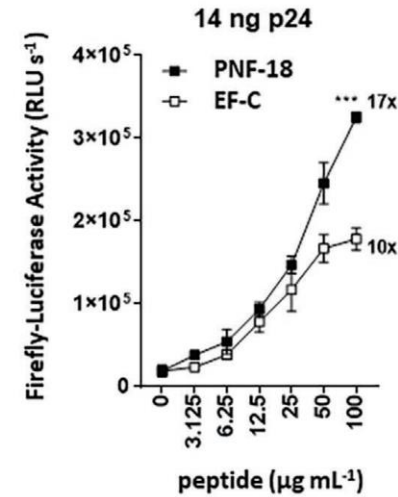
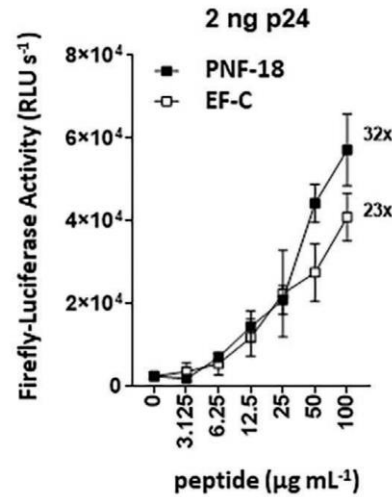
Head-to-Head Comparison



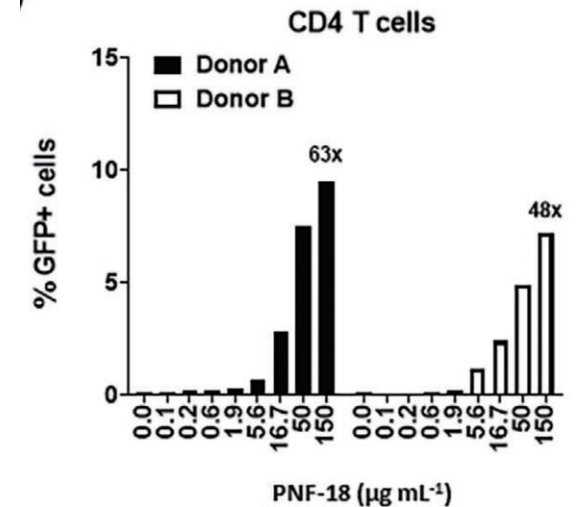
MAX PLANCK INSTITUTE
FOR POLYMER RESEARCH



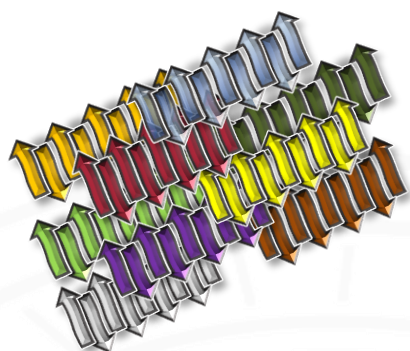
PNF-18 performs similar to or better than EF-C with different viral vectors and in various cells.



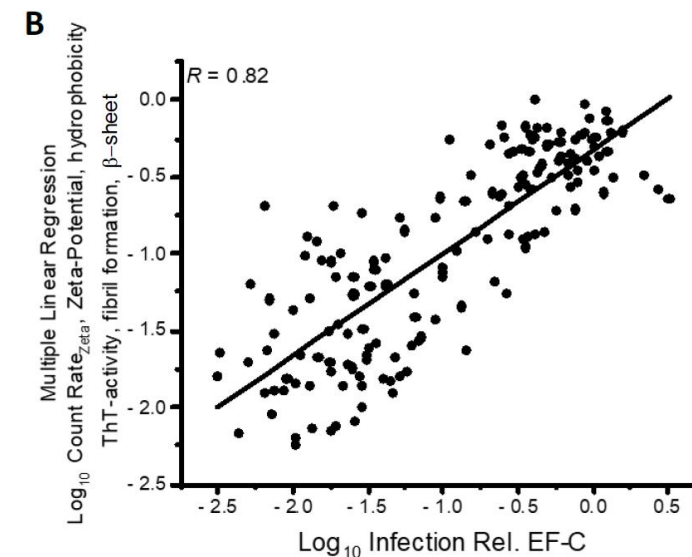
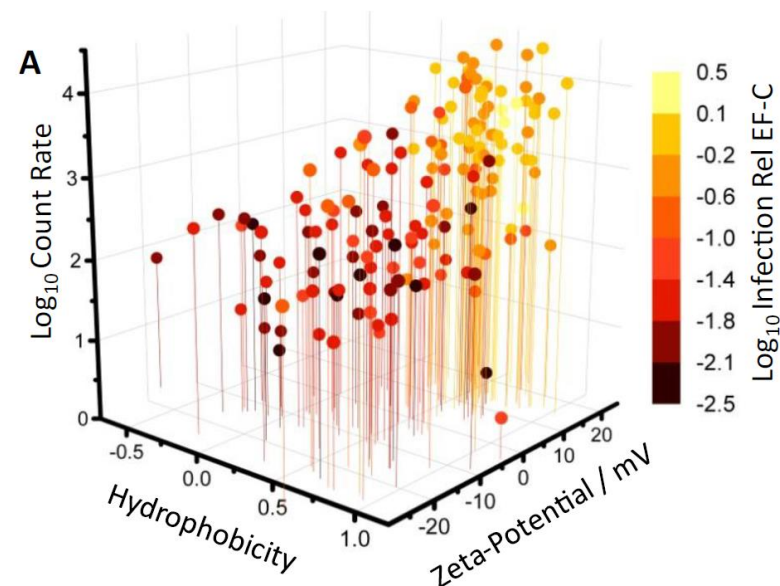
Lentiviral vector in HEK 293T cells



Peptide Library



> 160 peptides
Infectivity, TEM
morphology, Zeta
potential, ThT activity,
IR spectra,
Aggregation,



A combination of descriptors correlate with high activity
→ Multiple linear regression



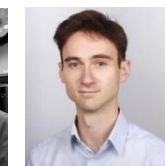
K. Kaygisiz



L. Rauch

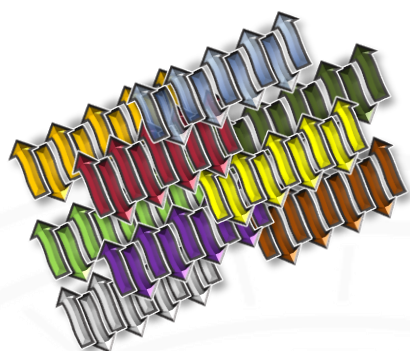


A. Dutta



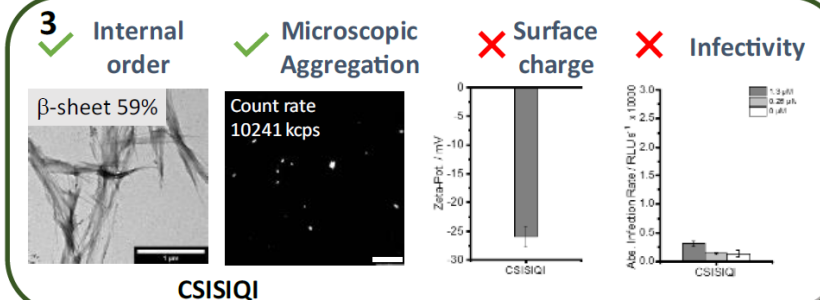
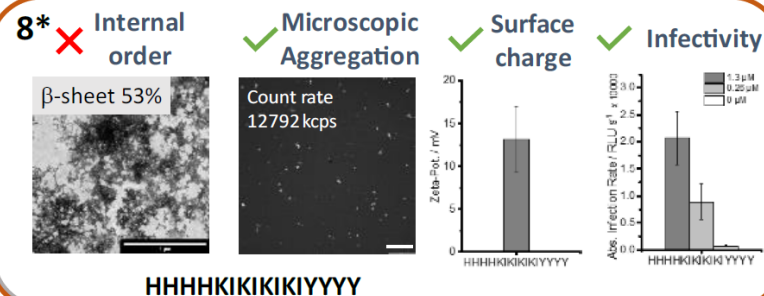
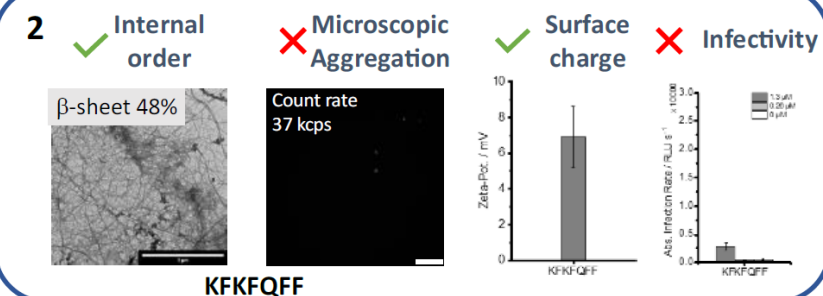
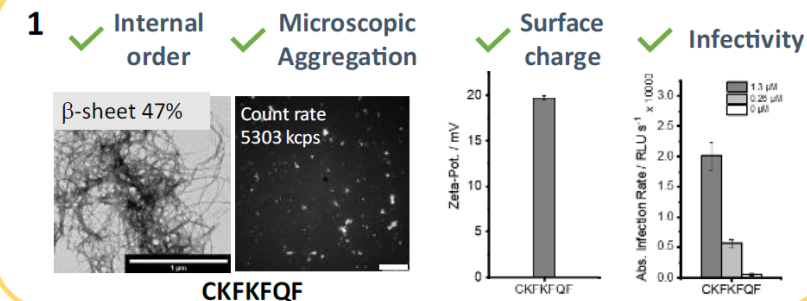
T. Bereau

Peptide Library

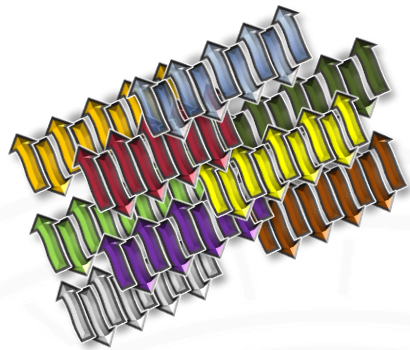


> 160 peptides
Infectivity, TEM
morphology, Zeta
potential, ThT activity,
Aggregation,

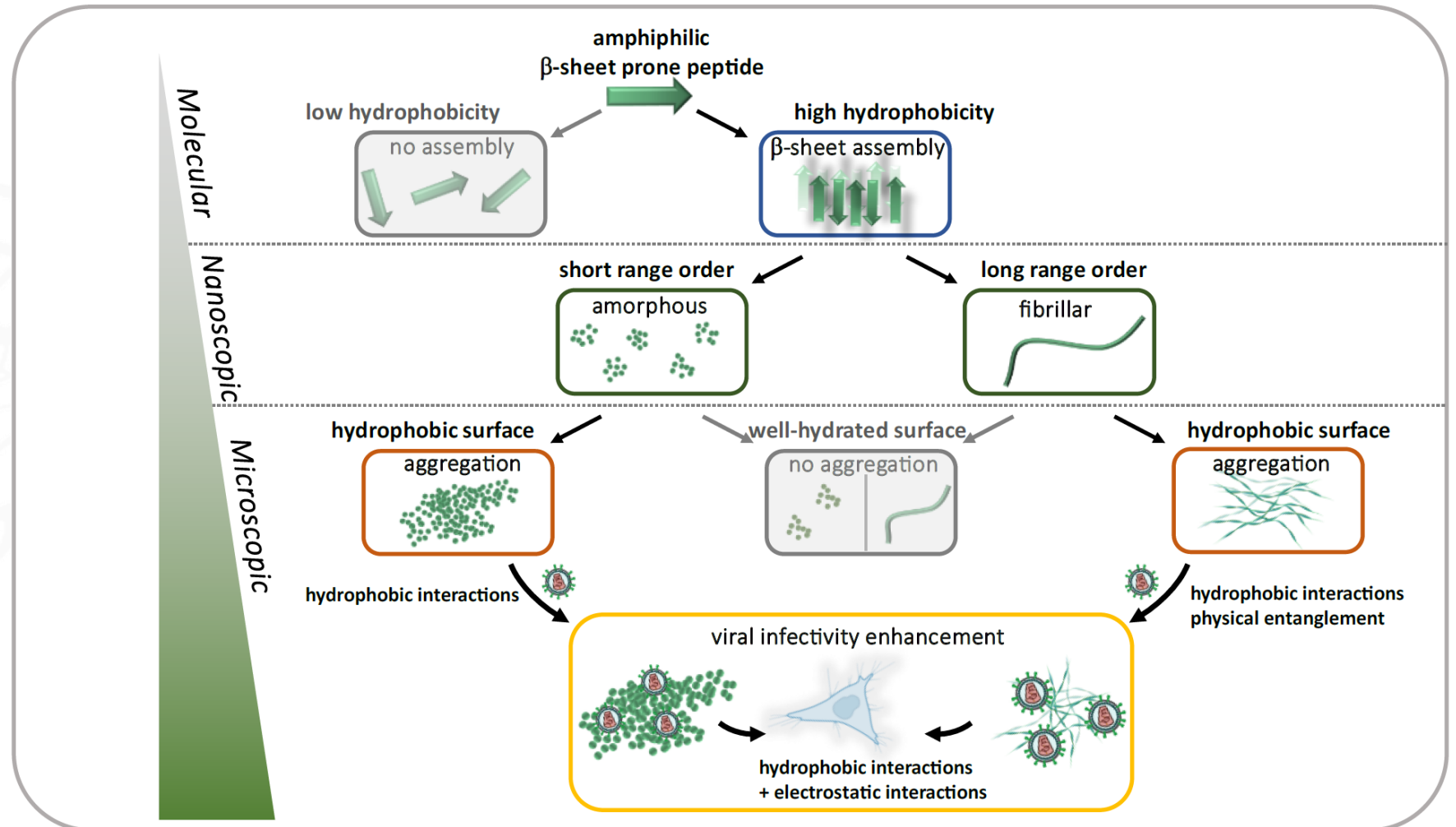
	Internal order	Microscopic Aggregation	Surface charge	Infectivity
1	✓	✓	✓	✓
2	✓	✗	✓	✗
3	✓	✓	✗	✗
4	✗	✓	✗	✗
5	✗	✗	✓	✗
6	✓	✗	✗	✗
7	✗	✗	✗	✗
8*	✗	✓	✓	✓



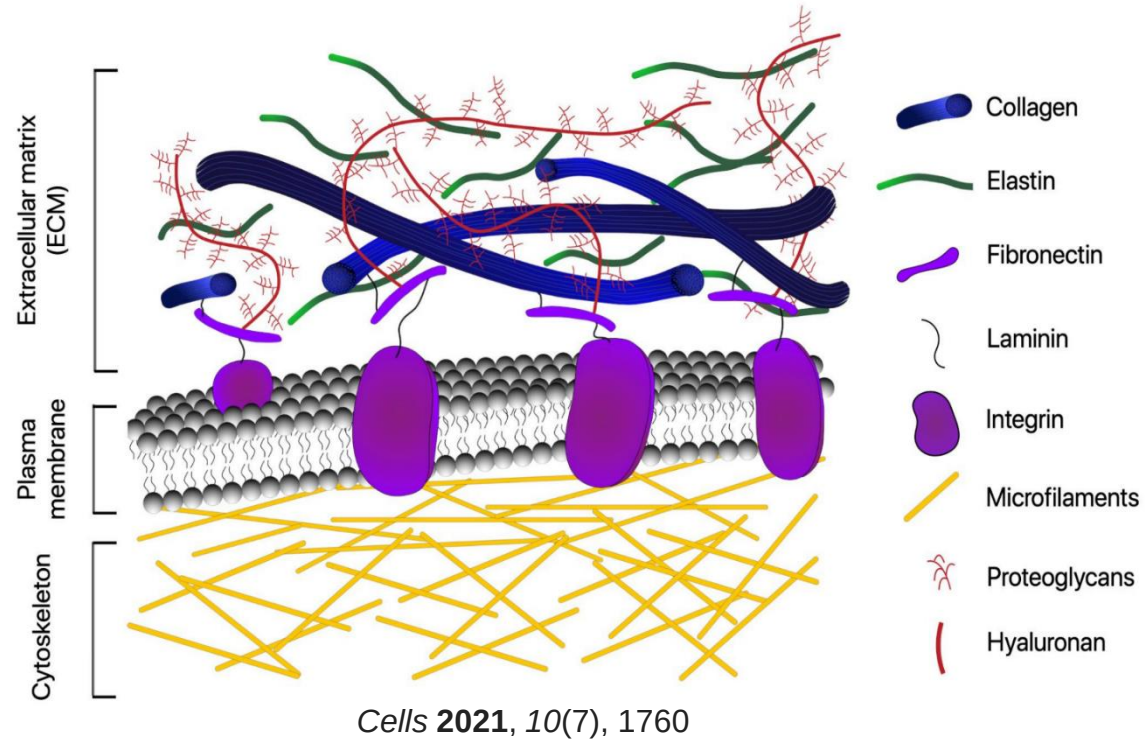
Peptide Library



> 160 peptides
Infectivity, TEM
morphology, Zeta
potential, ThT activity,
IR spectra,
Aggregation,



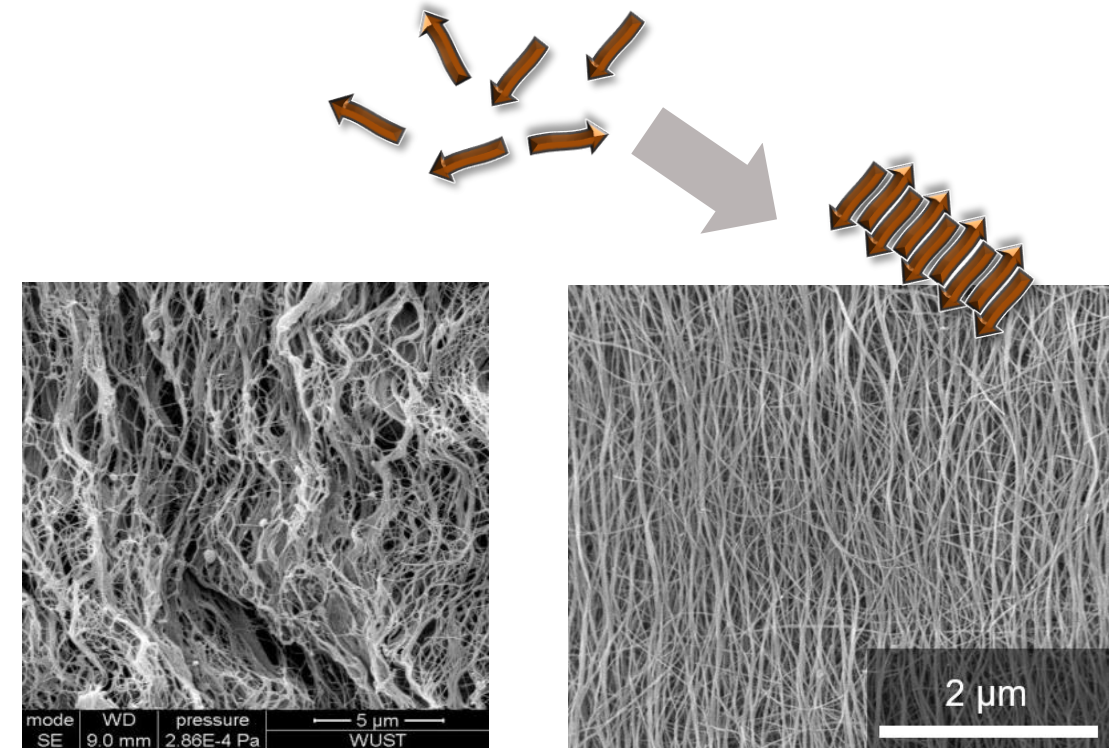
Extracellular Matrix (of Animals)



Function of the ECM

- Topographical cues
- Mechanical support
- Bioactive signals
- ...

Self-Assembled Peptide Nanofibers



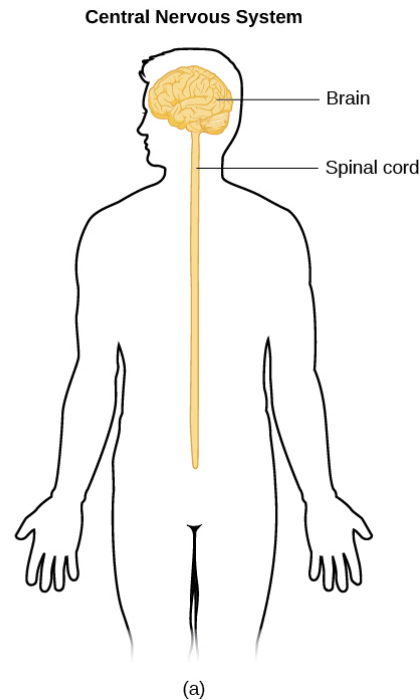
Decellularized heart
valve tissue

Injuries of the Nervous System

Central Nervous System (CNS) Brain + Spinal Cord

Injuries / Diseases

- Traumatic injuries to brain or spinal cord
- Tumors
- Surgical interventions
- etc.



Peripheral Nervous System (PNS) All Nerves outside the CNS

Injuries

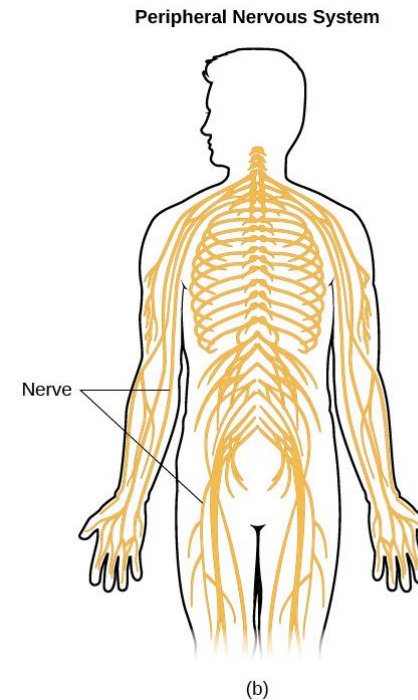
Mainly traumatic injuries of face and extremities:
Work-, Traffic-, Sports-Accidents
War related injuries

Effects

- Loss of sensory function
- Loss of motor control

Prevalence

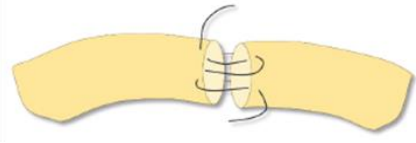
- ca. 350.000 / Year in USA
- ca. 300.000 / Year in EU



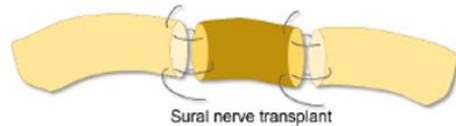
Current Treatment Methods for PNS Injuries

Surgical Interventions

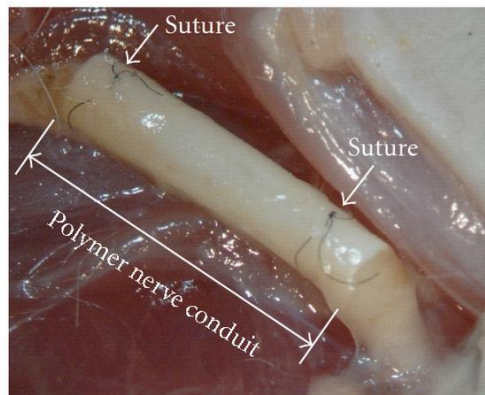
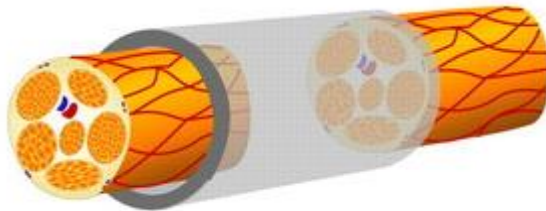
A) Direct Suturing



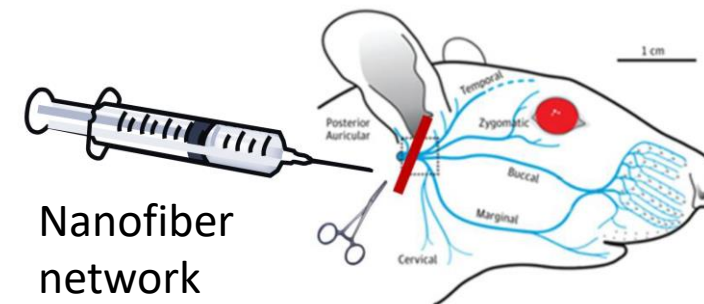
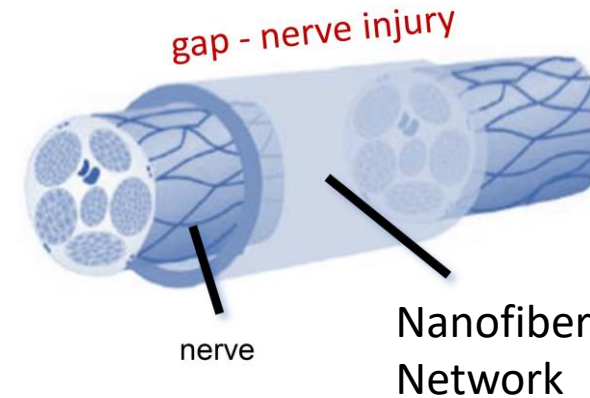
B) Autograft Transplant



C) Nerve guidance conduit



D) Local injection of nanofiber network



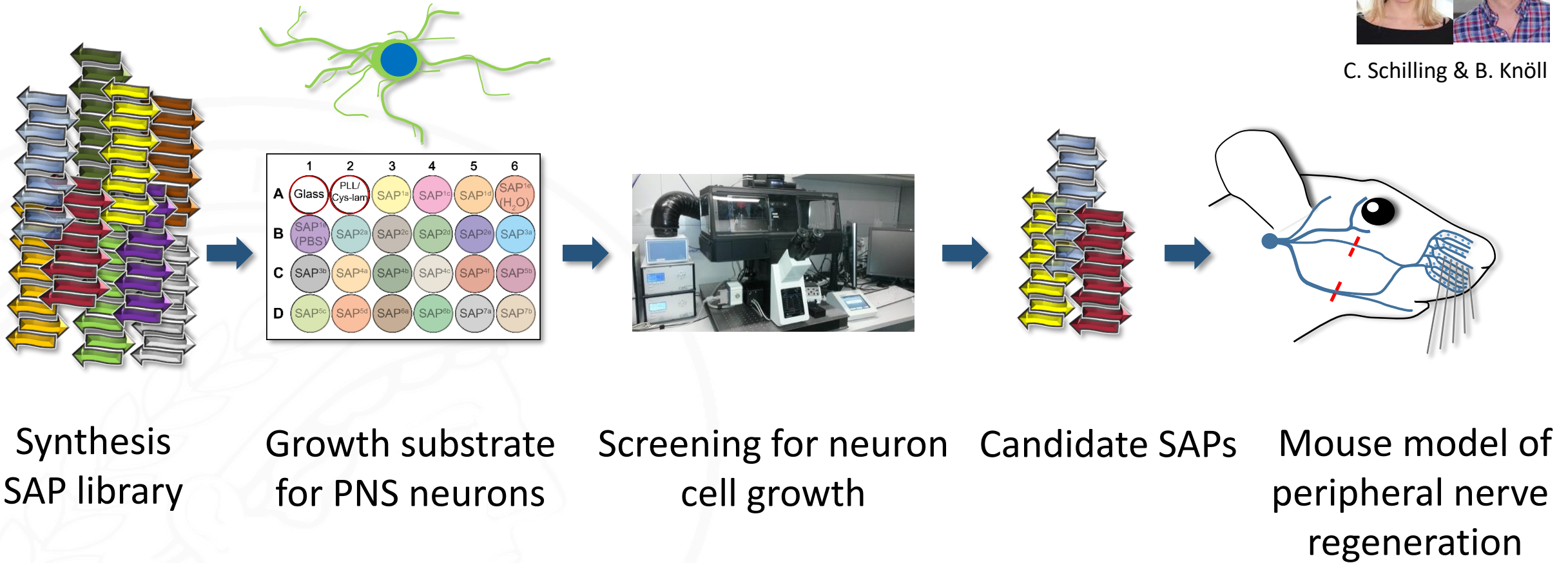
Bioactive Scaffolds for Nerve Regeneration



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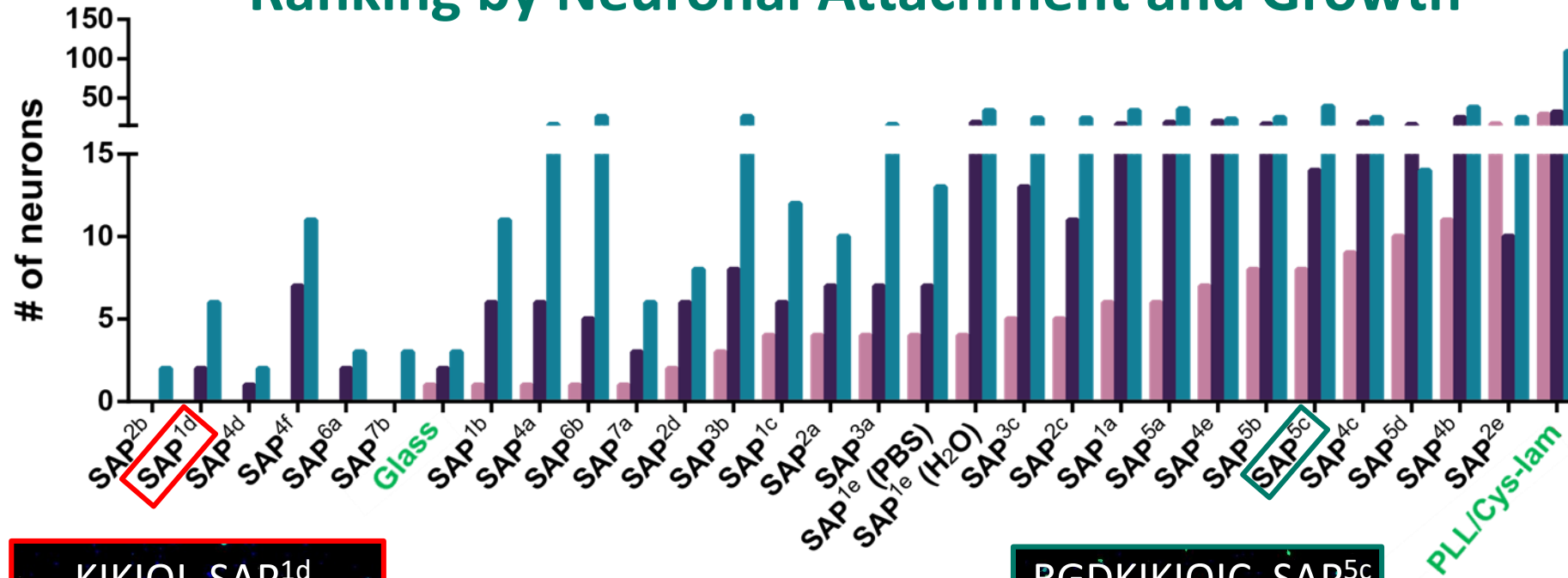


C. Schilling & B. Knöll

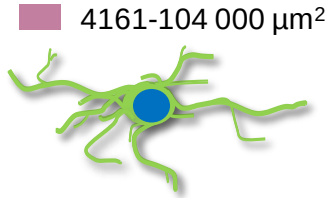


Bioactivity: Peptide Nanofibers as Nerve Cell Scaffolds

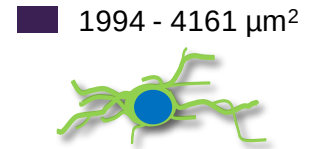
Ranking by Neuronal Attachment and Growth



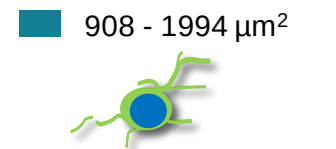
Large neurons



Medium neurons



Small neurons



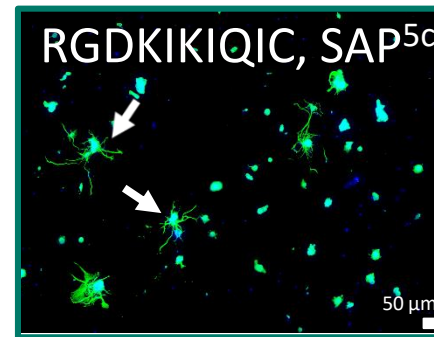
C. Schilling & B. Knöll

KIKIQI, SAP^{1d}



No growth

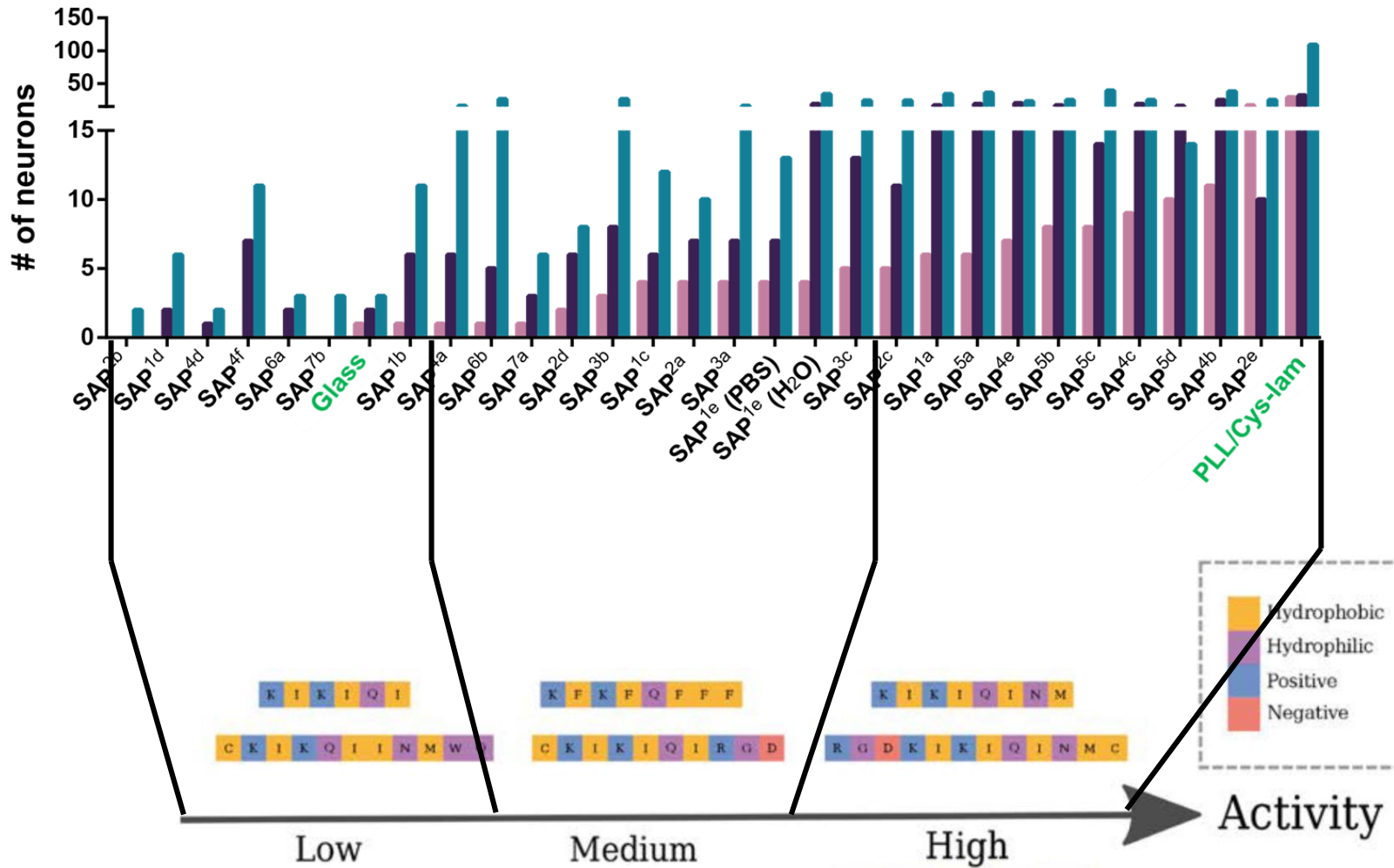
RGDKIKIQIC, SAP^{5c}



Growth stimulation



A. Dutta, T. Berau
(MPIP / U. Amsterdam)



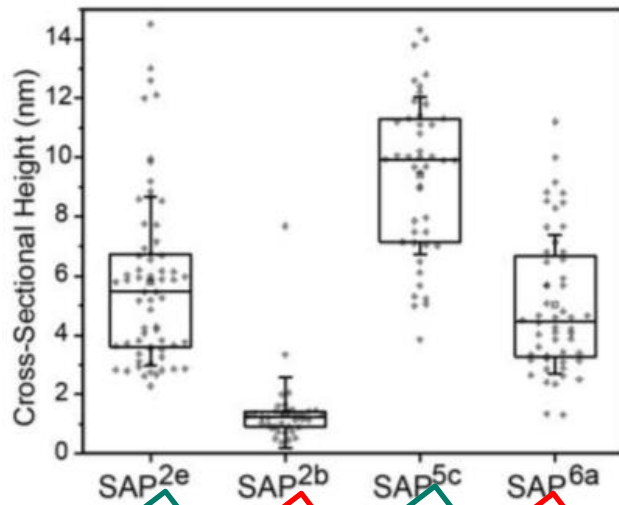
Pattern Analysis

- Coarse Graining of sequences
- Pattern analysis in high and low activity samples
- Alternating hydrophobic and positively charged

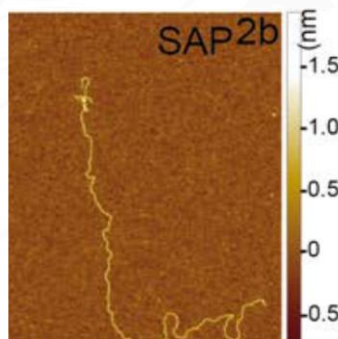


Correlating Structure with Activity

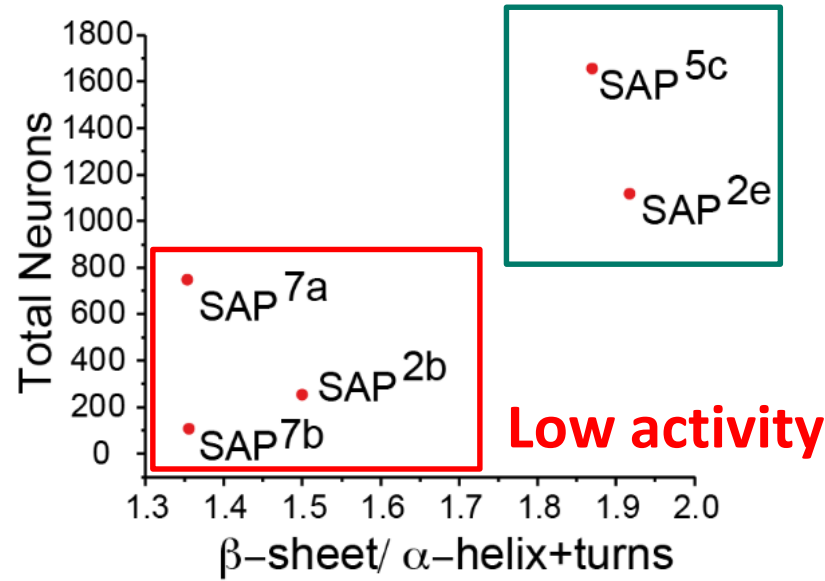
Microscopy (AFM):



CKFKFQF
KFKFQF
RGDKIKIQIC
KIKIQIRGD

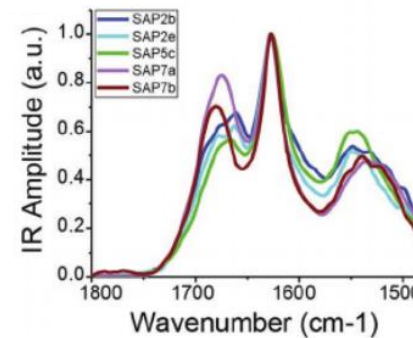


Spectroscopy (FTIR):



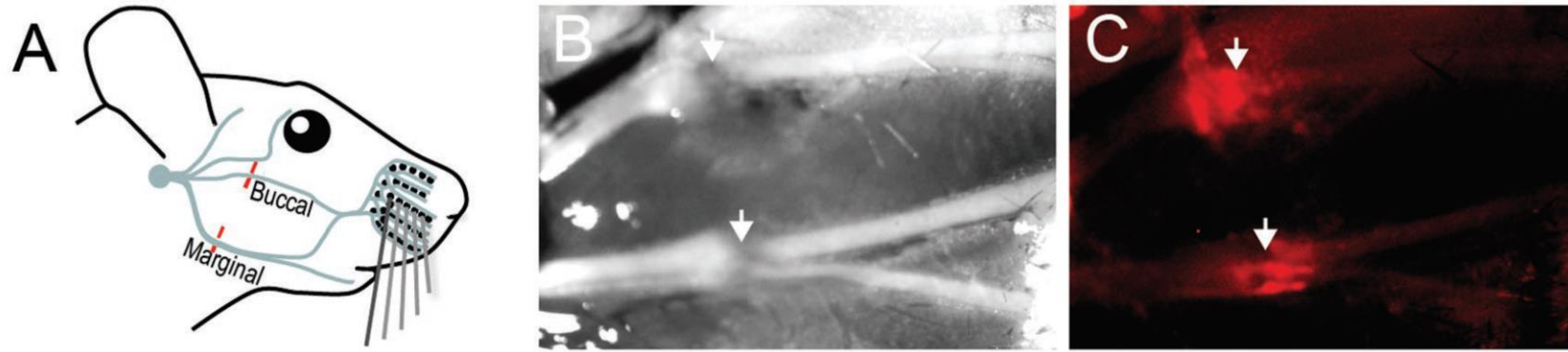
High neuronal activity

- Positive charges
- Thicker fibers
- Higher β -sheet content



S. Ruggeri & T. Knowles

Peptide Nanofibers Support Peripheral Nerve Regeneration



in vivo :

- Surgical lesion of facial nerve controlling whisker movement
- Injection of Nanofibers at lesion site
- Material remains at lesion site for at least 3 weeks



Peptide Nanofibers Support Peripheral Nerve Regeneration

Day 0



Day 18

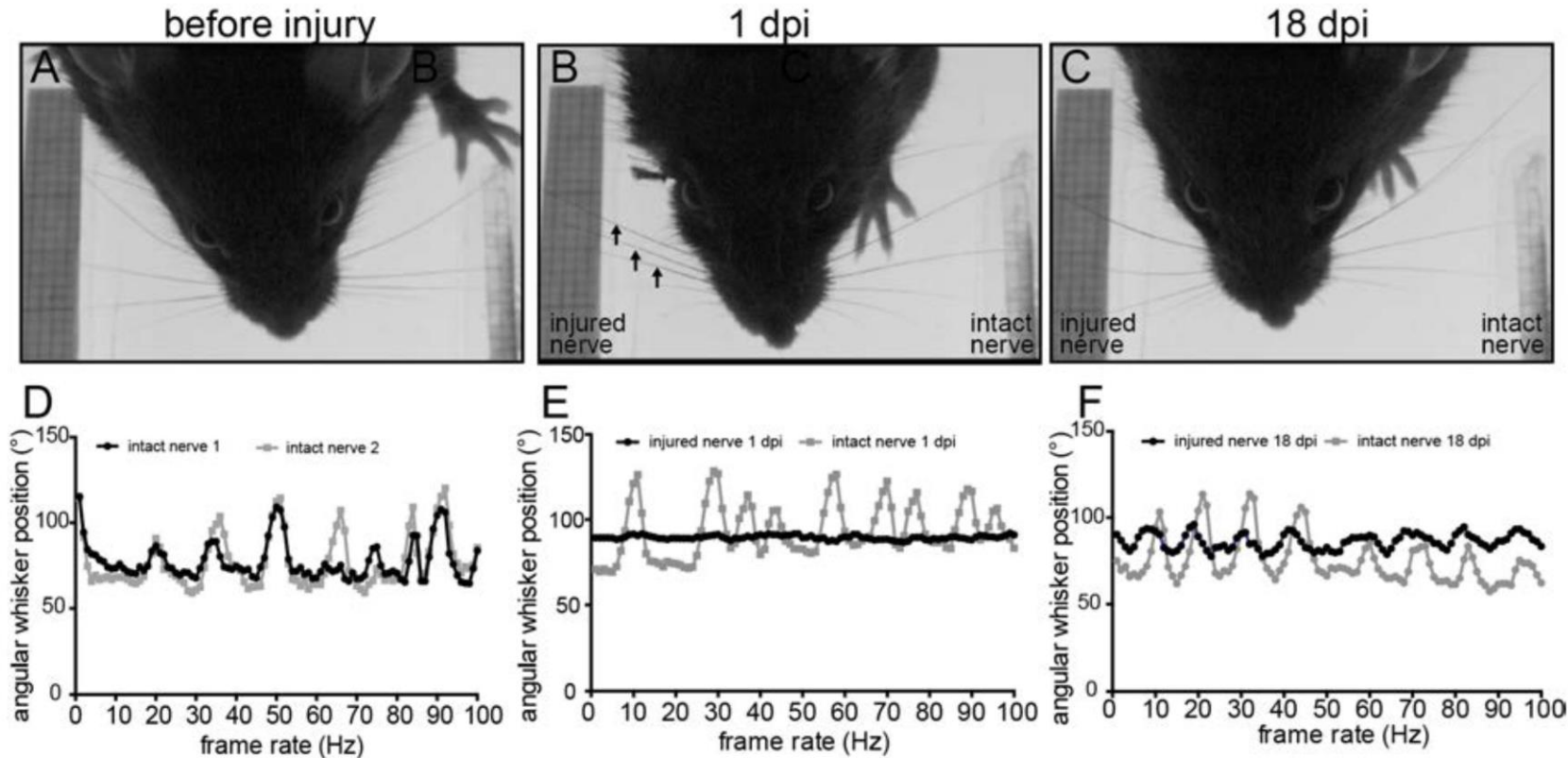


in vivo :

- Surgical lesion of facial nerve controlling whisker movement
- Injection of Nanofibers at lesion site
- Material remains at lesion site for at least 3 weeks



Peptide Nanofibers Support Peripheral Nerve Regeneration



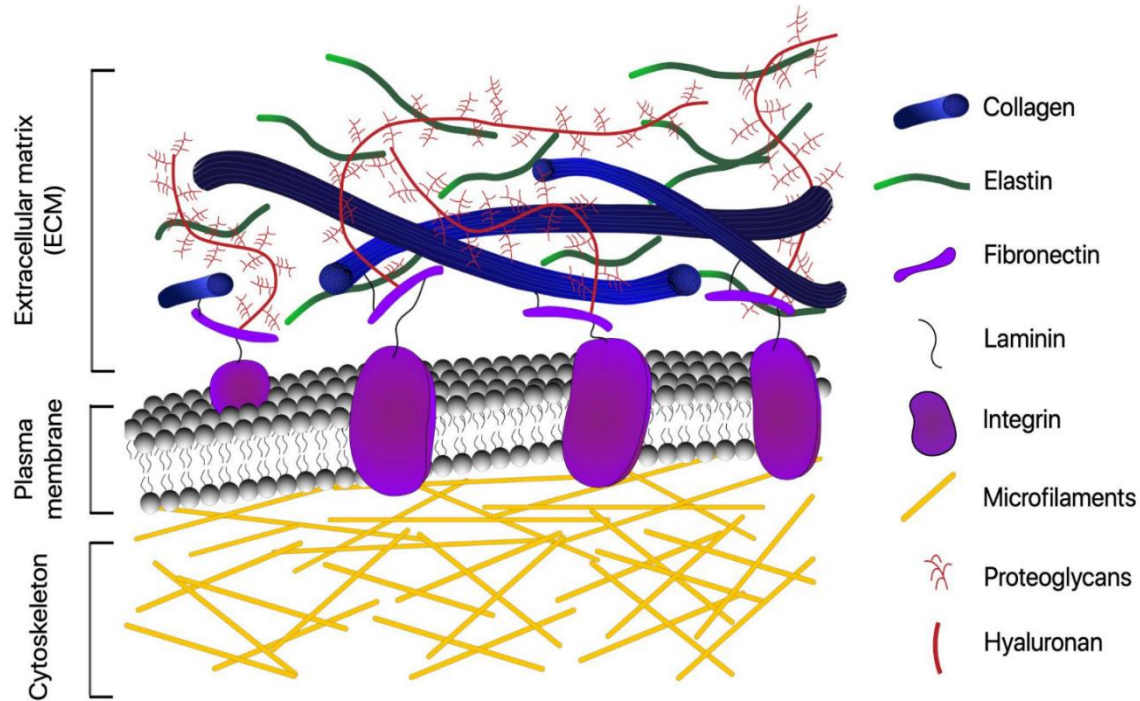
in vivo :

- Surgical lesion of facial nerve controlling whisker movement
- Injection of Nanofibers at lesion site
- Material remains at lesion site for at least 3 weeks
- Partial recovery 18 days post injection



Mimicking the ECM in Space and Time

Extracellular Matrix (of Animals)



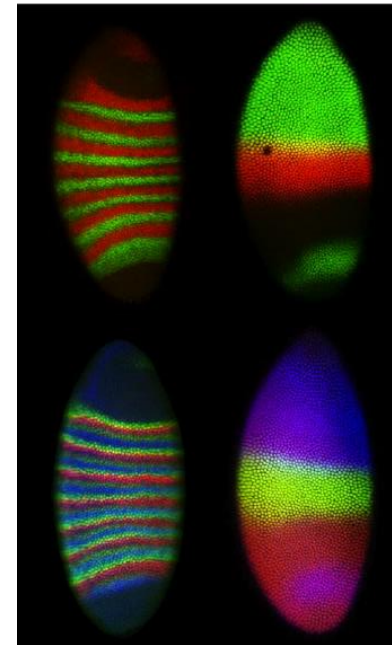
Cells **2021**, 10(7), 1760

Function of the ECM

- Topographical cues
- Mechanical support
- Bioactive signals
- ...

Biology is Dynamic and Non-Homogeneous

- Constant Remodeling
- Spatio-temporal control over signals



Patterns of protein expression
during fruit fly development

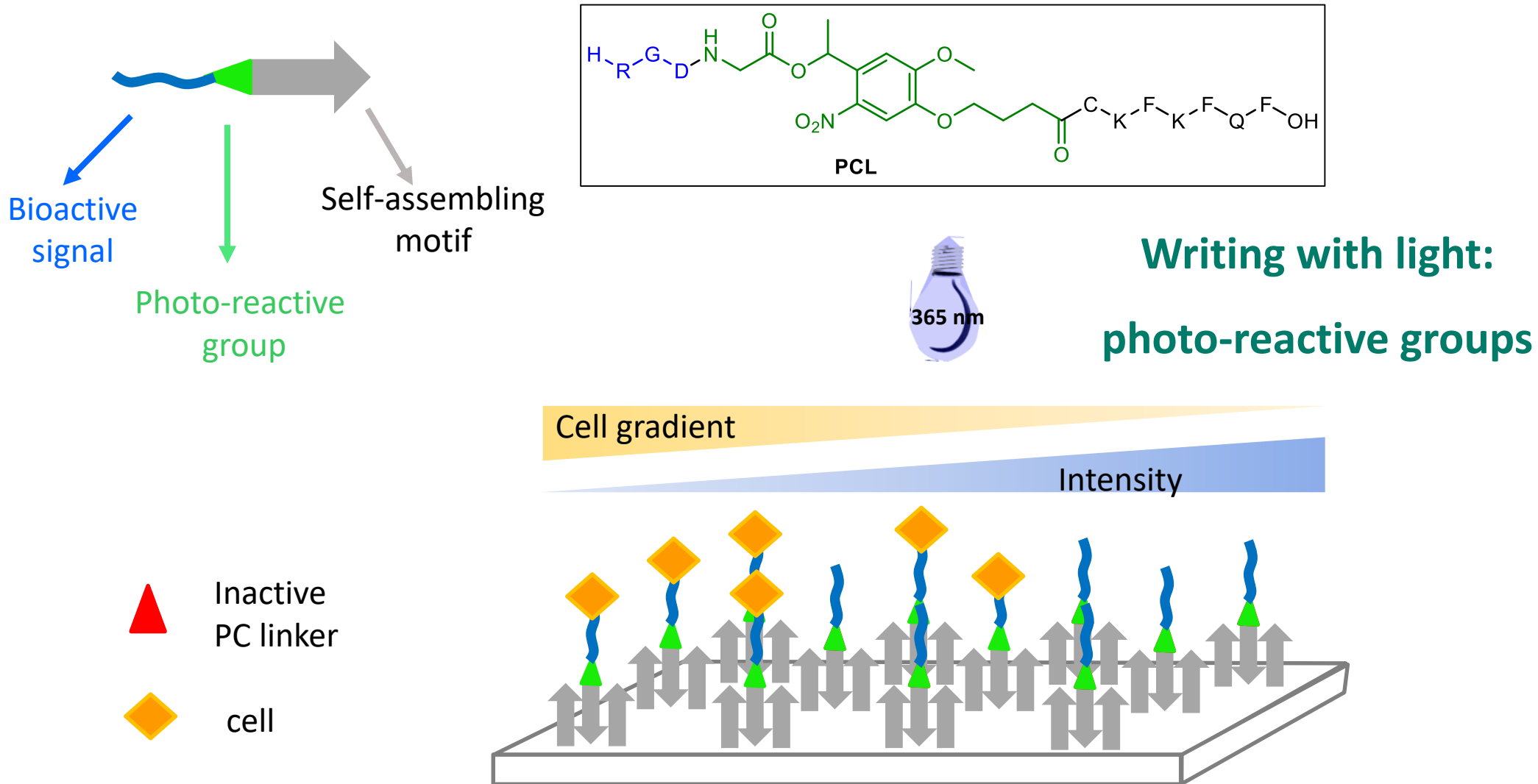


Biological Gradients:

Migration
Polarization
Differentiation

...

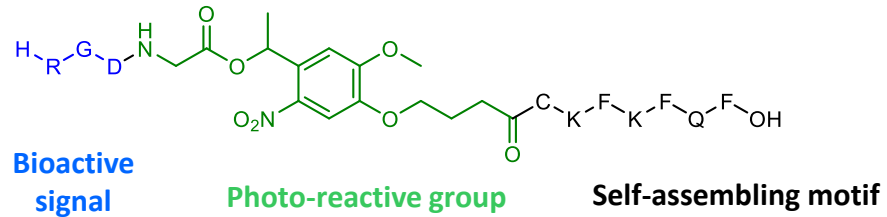
Spatio-Temporal Control over Bioactivity



A. Ender

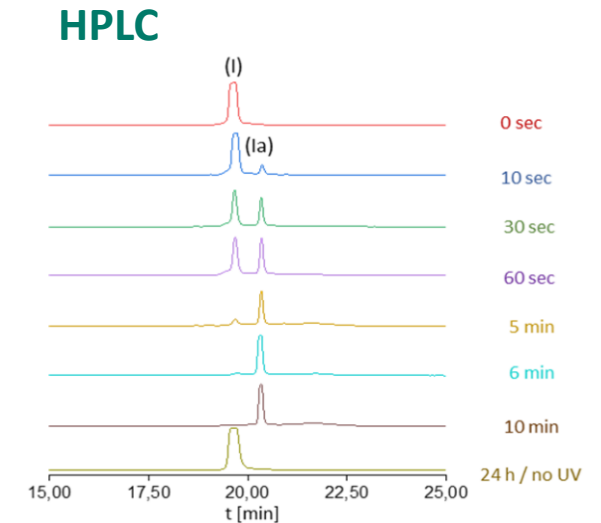
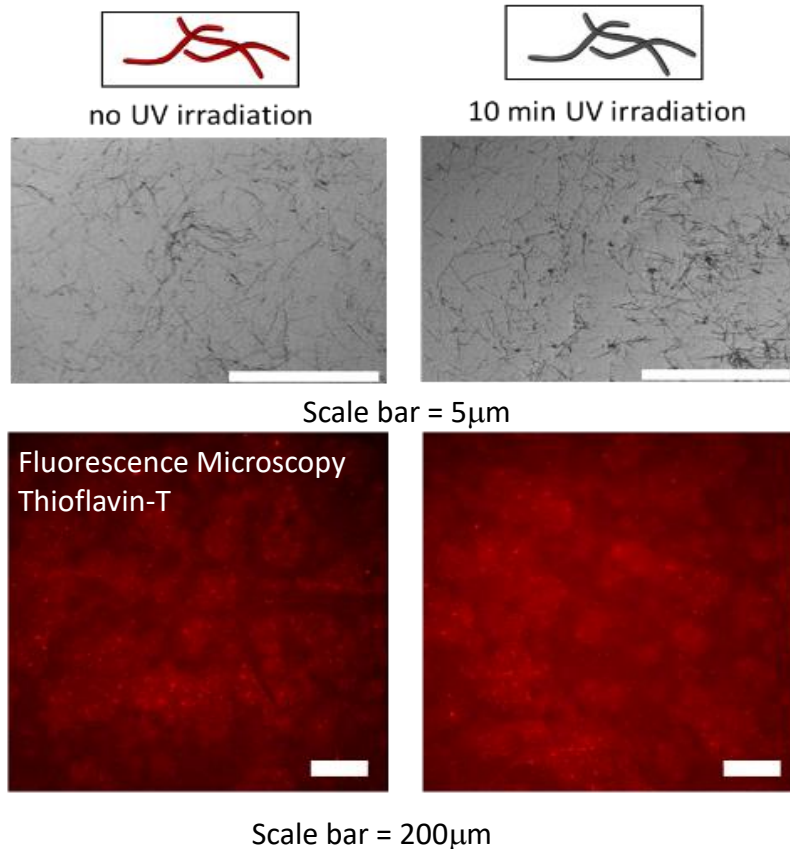


K. Kaygisiz



UV Exposure:

- Fiber structure and amyloid character remains intact
- Cleavage within minutes of UV exposure

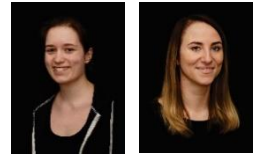


A. Ender & K. Kaygisiz

Creating Cell-Adhesive Gradients on Surfaces



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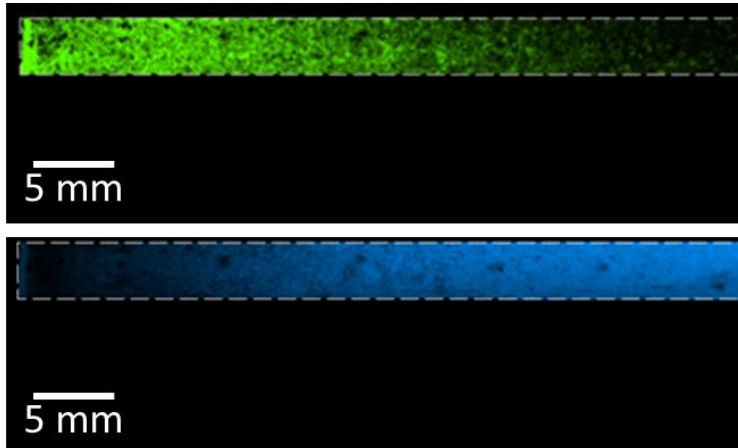
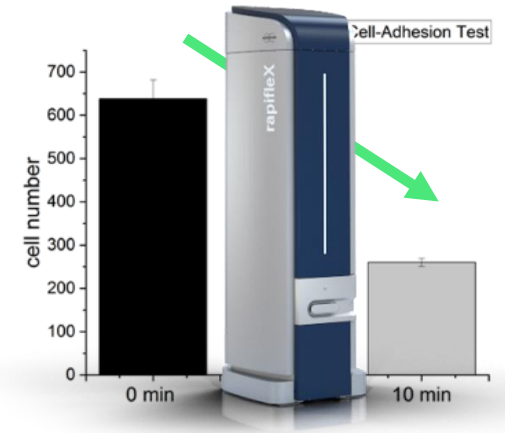
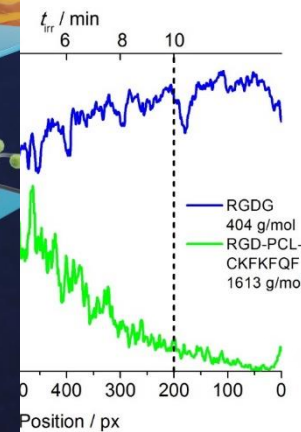
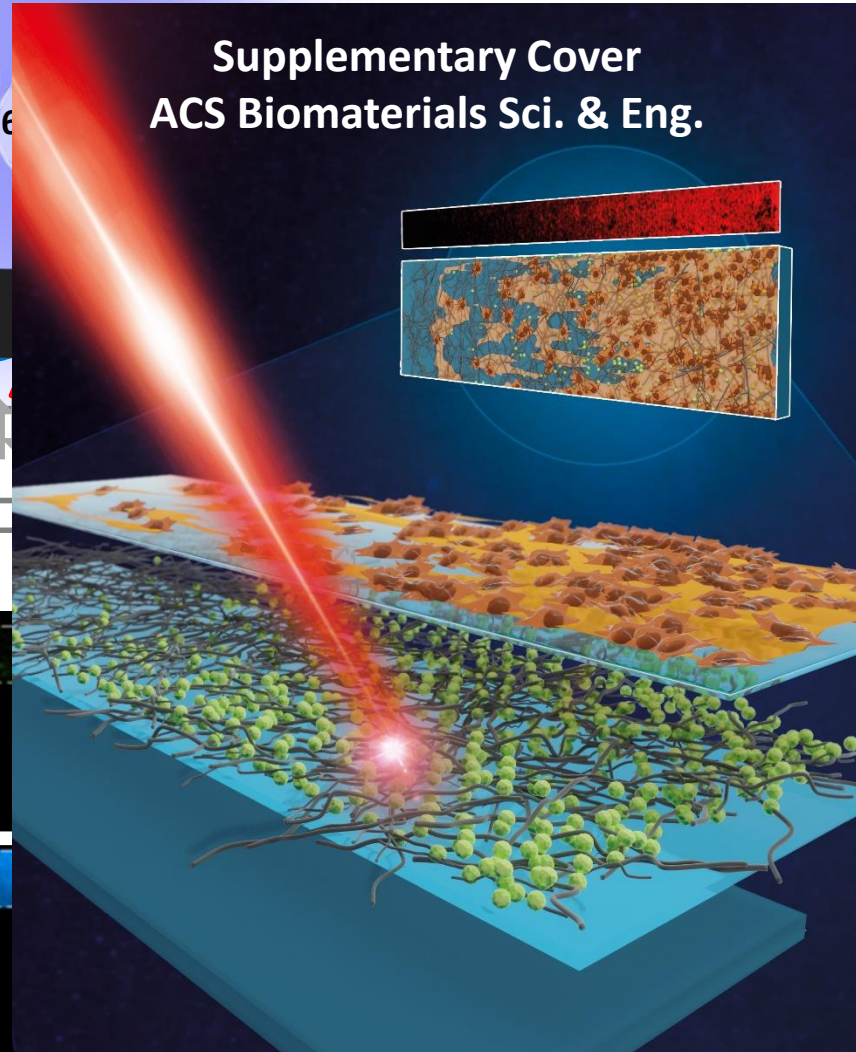
Supplementary Cover
ACS Biomaterials Sci. & Eng.

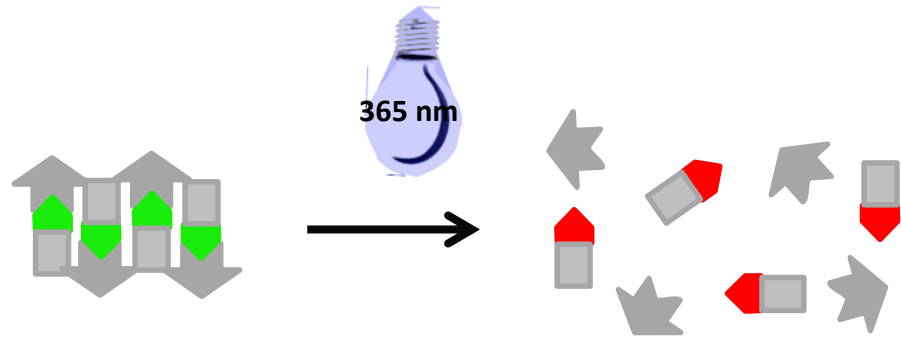
on surfaces

re to cleave off epitope

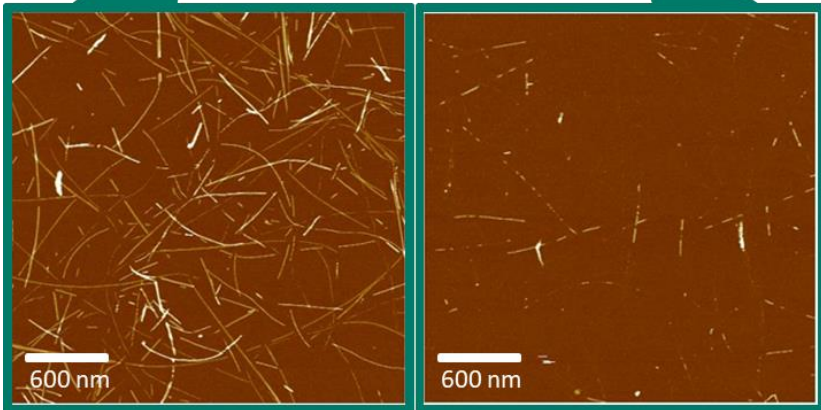
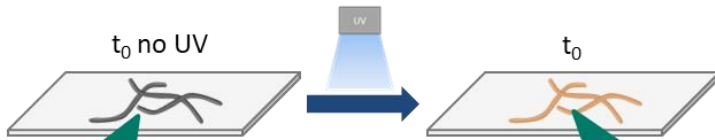
firms gradients

llular adhesion



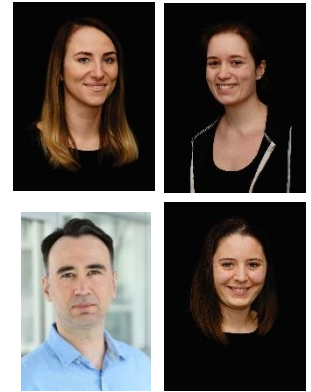


Liquid AFM



Patterning of Bioactive Fibers

- Cleavable assembly motif
- Photoirradiation disassembles fibers
- Cells only attach on intact fibers

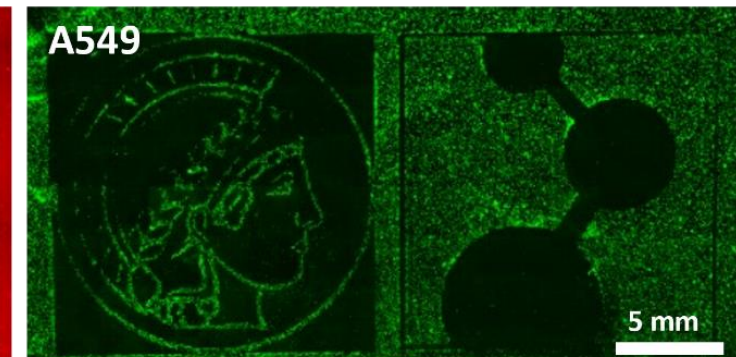


A. Ender, A. Kaczmarek,
D. Koutsouras, K. Kaygisiz

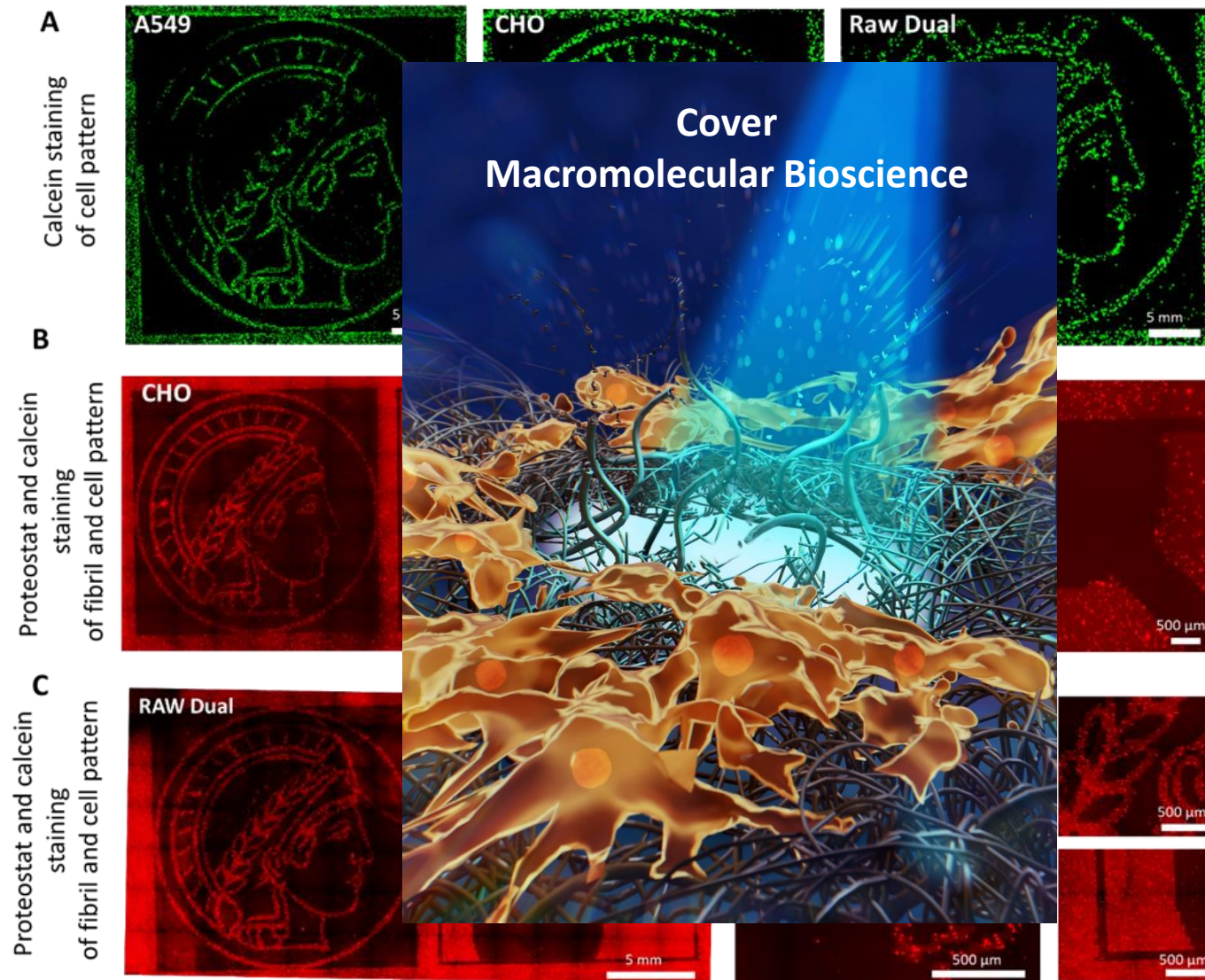
Photopatterning



Cell Patterning



Defined Cell Patterns



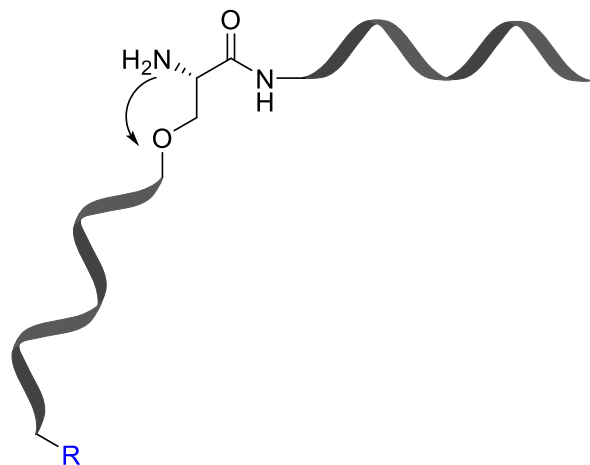
pH-induced Assembly of Peptide Nanofibrils



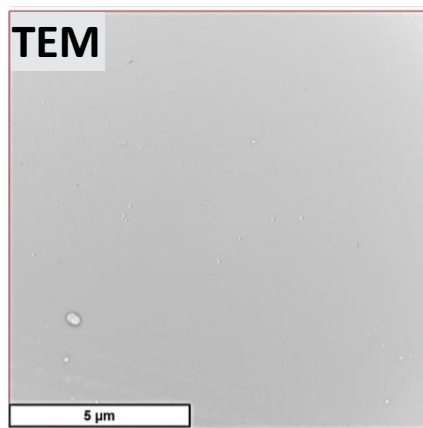
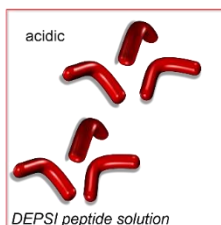
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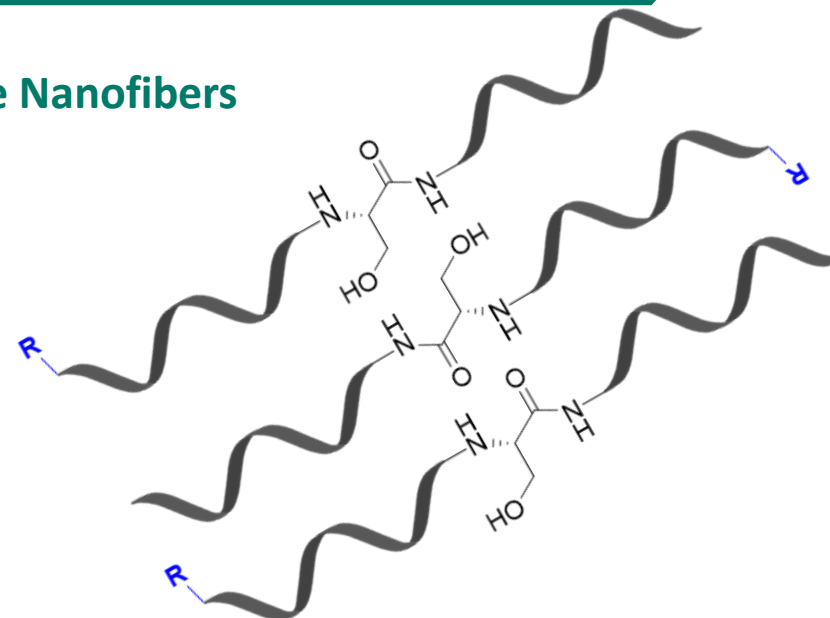
Depsi Peptide



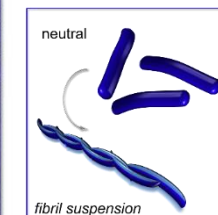
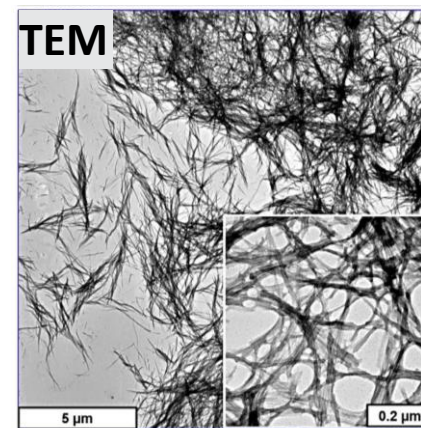
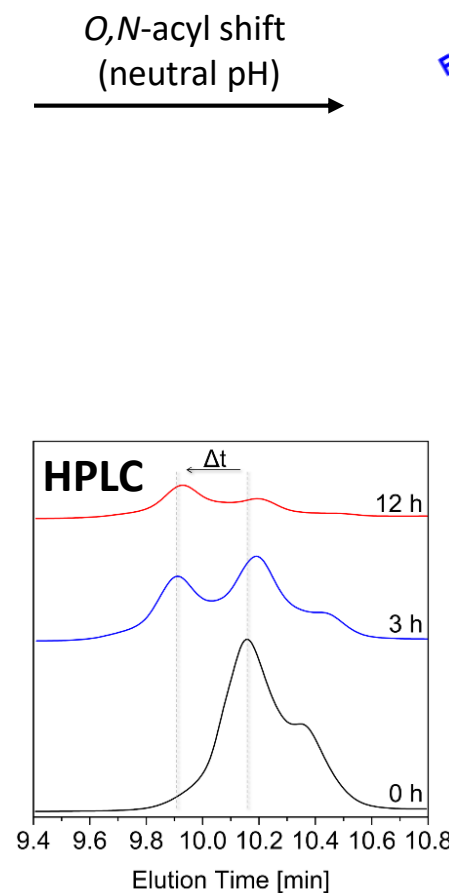
**No assembly in
acidic conditions**



Peptide Nanofibers

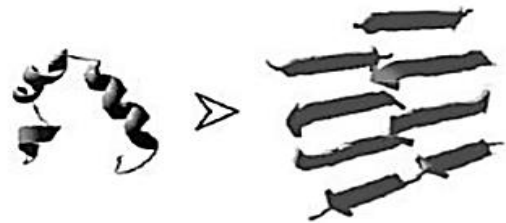


**Assembly at
neutral pH**

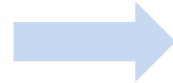


3D Systems: Peptide-Biopolymer Hybrids

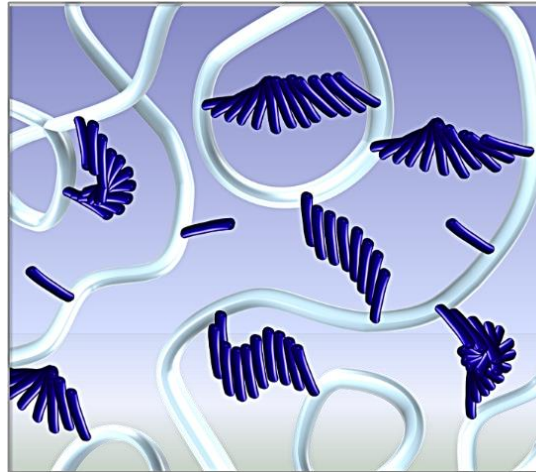
Supramolecular Gelator



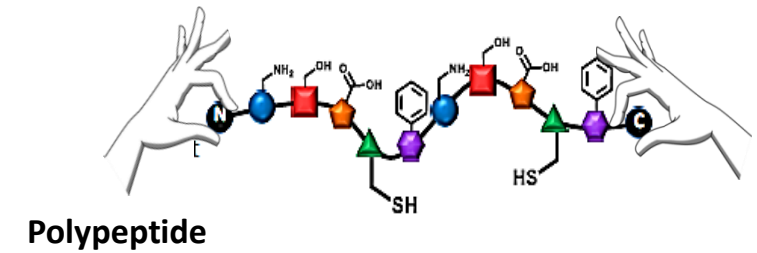
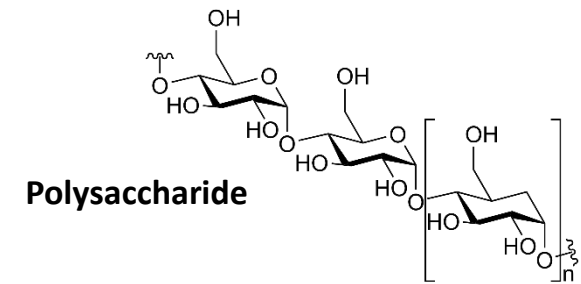
Peptide



Hybrid Hydrogel



Biopolymer Backbone

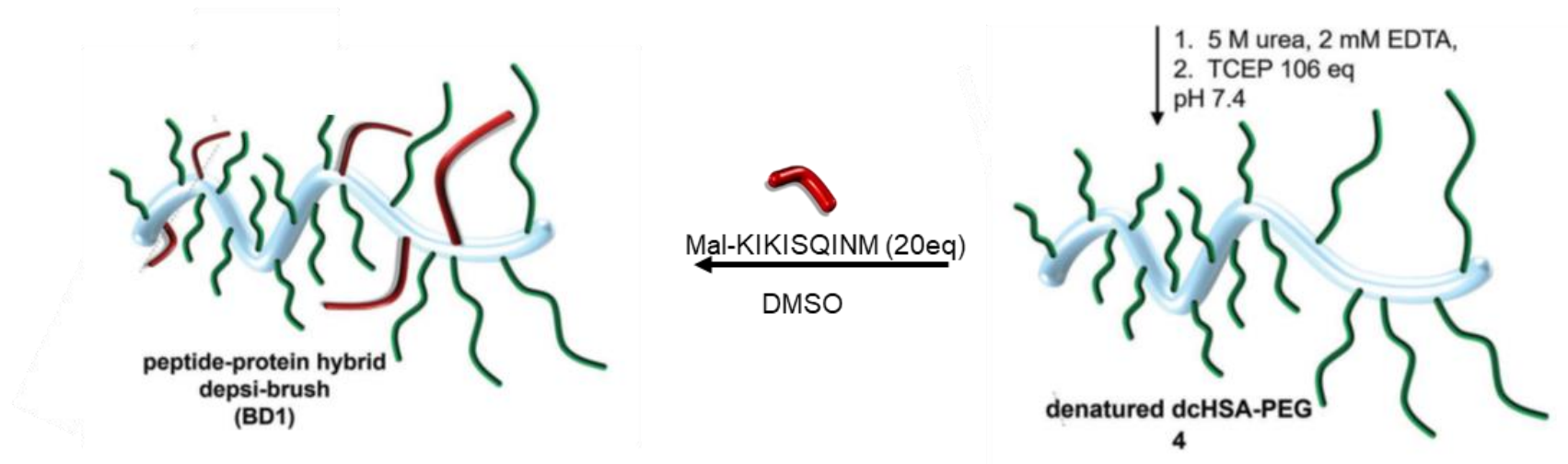
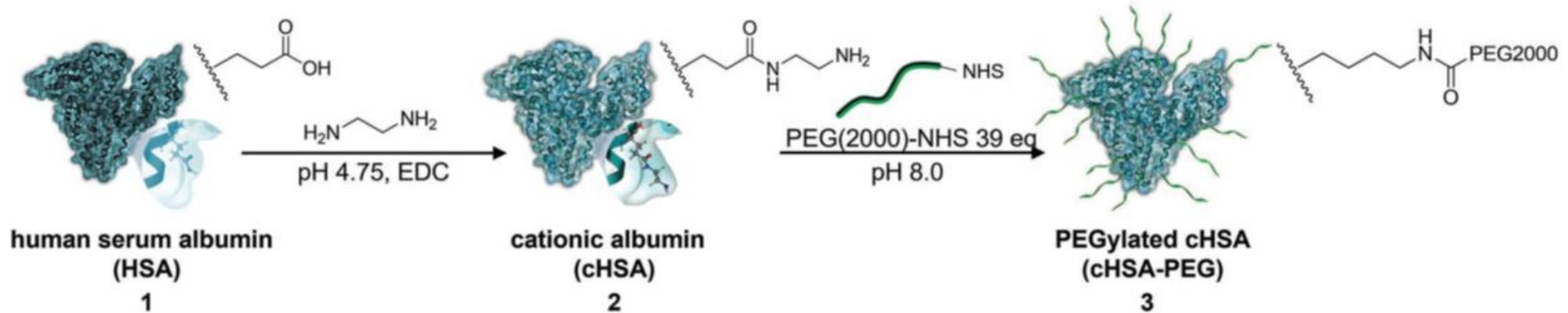


Nanofiber

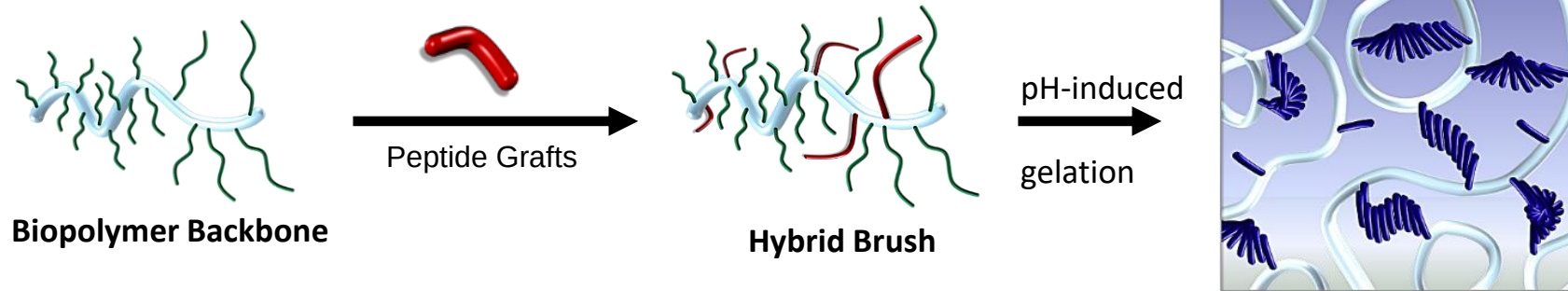
Hybrid Brush Synthesis



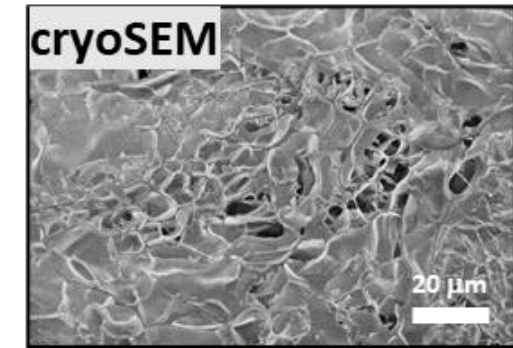
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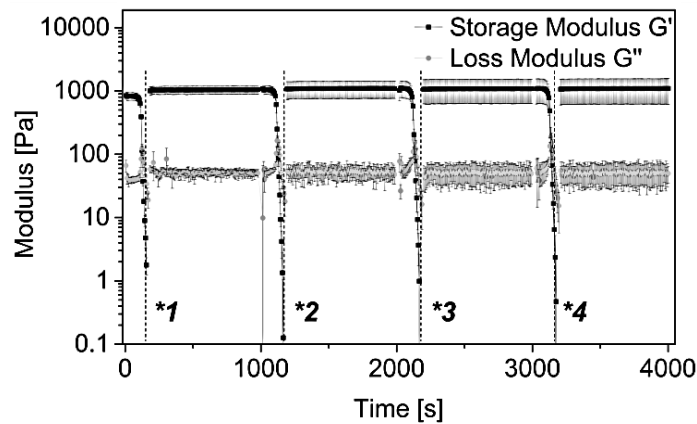
Thixotropic Hydrogels with Fast Recovery



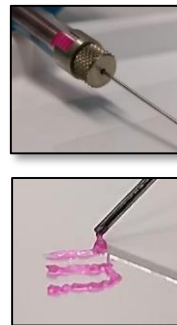
Porous Structure



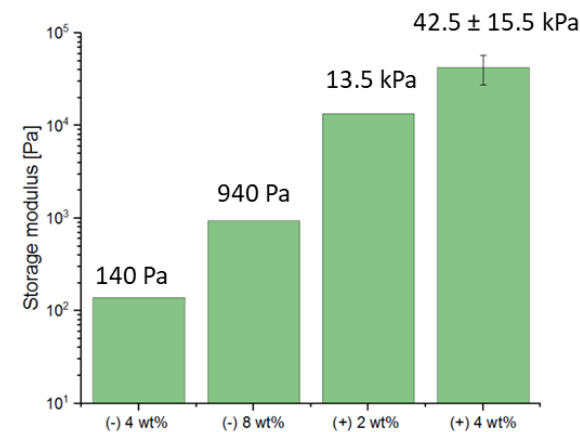
Fast, Quantitative Recovery



Injectable



Adjustable Stiffness



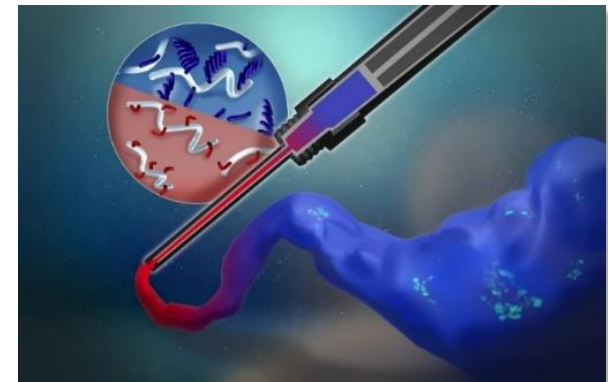
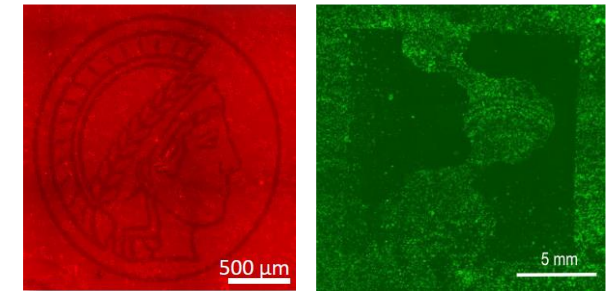
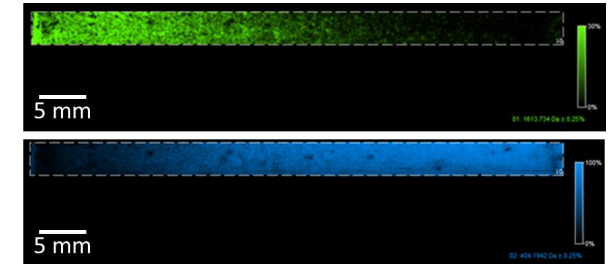
Additional Features

- Multiple backbones and peptides
- Enzymatically degradable
- Supports multiple cell types

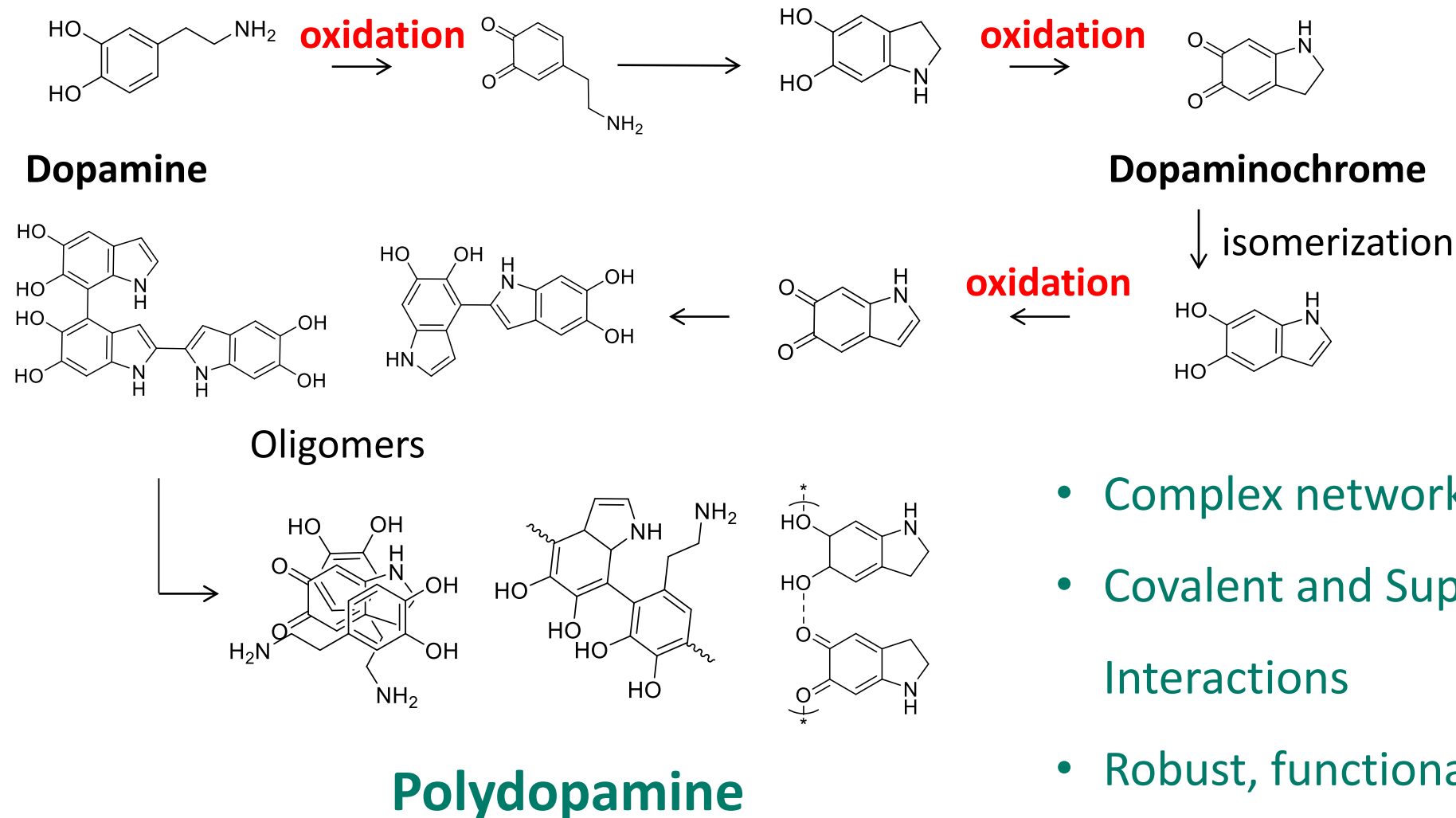
Industry Cooperation

CLARIANT 

- Optimizing Bioactivity
Viral Transduction
- Data Mining and Machine Learning
Property-Activity Correlation
Prediction of Functional Materials
- Spatial Control over Bioactivity
Patterns and Gradients
- Injectable, Self-Recovering Hydrogels
3D Cell Matrices and Rheology Modifiers

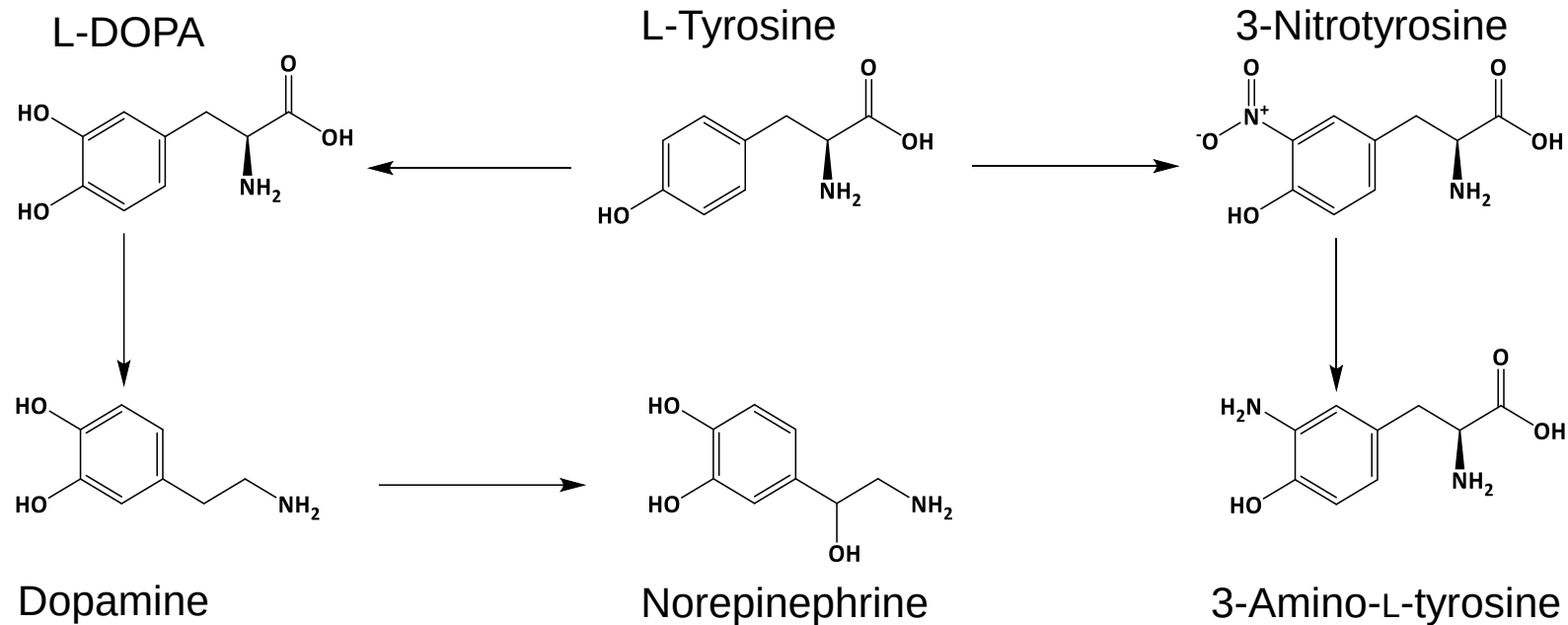


Polymers from Biogenic Self-Crosslinking Monomers

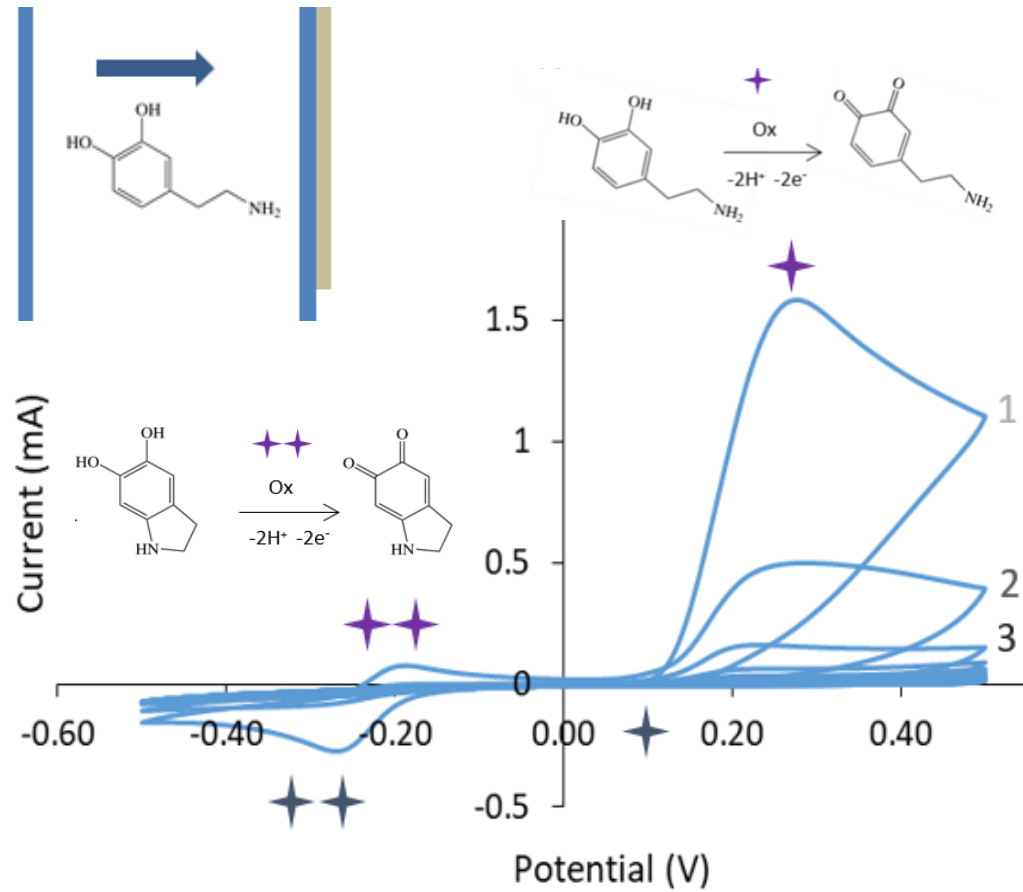


- Complex network
- Covalent and Supramolecular Interactions
- Robust, functional materials

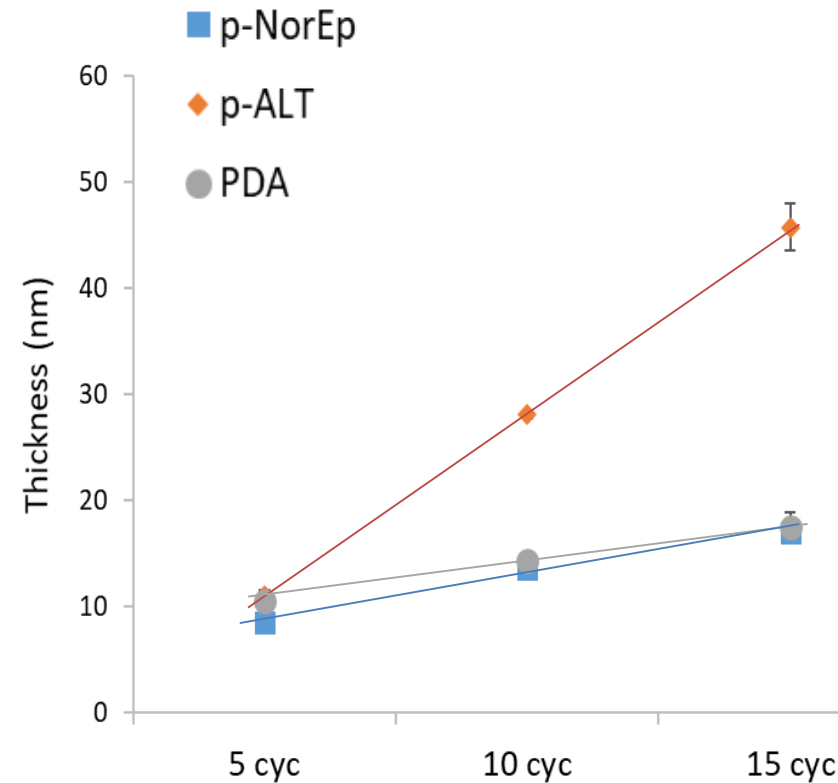
Electrodeposition of Biogenic Monomers



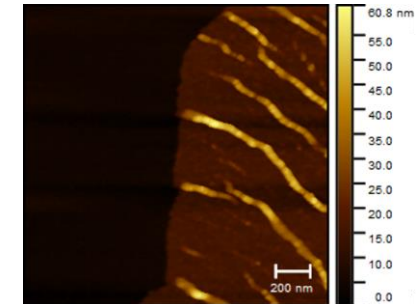
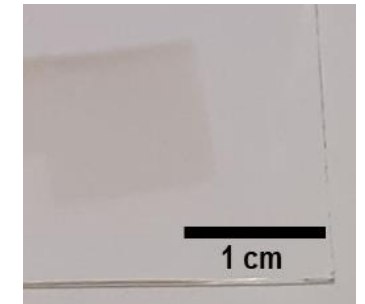
Electropolymerization



Linear film growth



Film Transfer

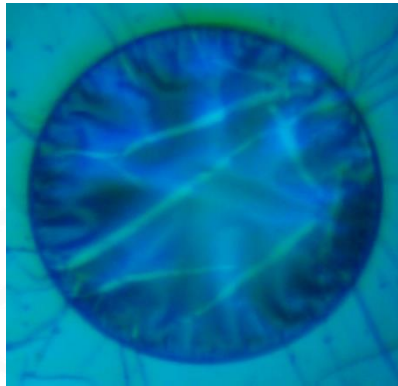


EU ITN „Borges“: Biosensors

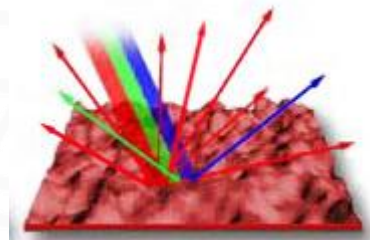
TRR 234: Switchable membranes for photocatalysis
 Actuators, Water Filtration Membranes

Photo-Actuating Thin Films

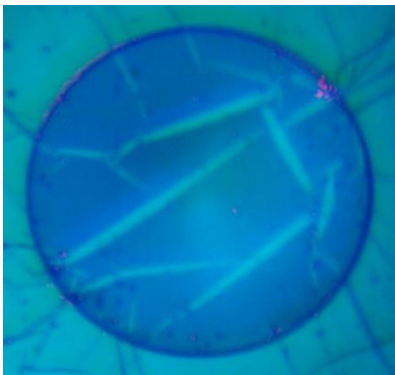
Laser OFF → Membrane wrinkled



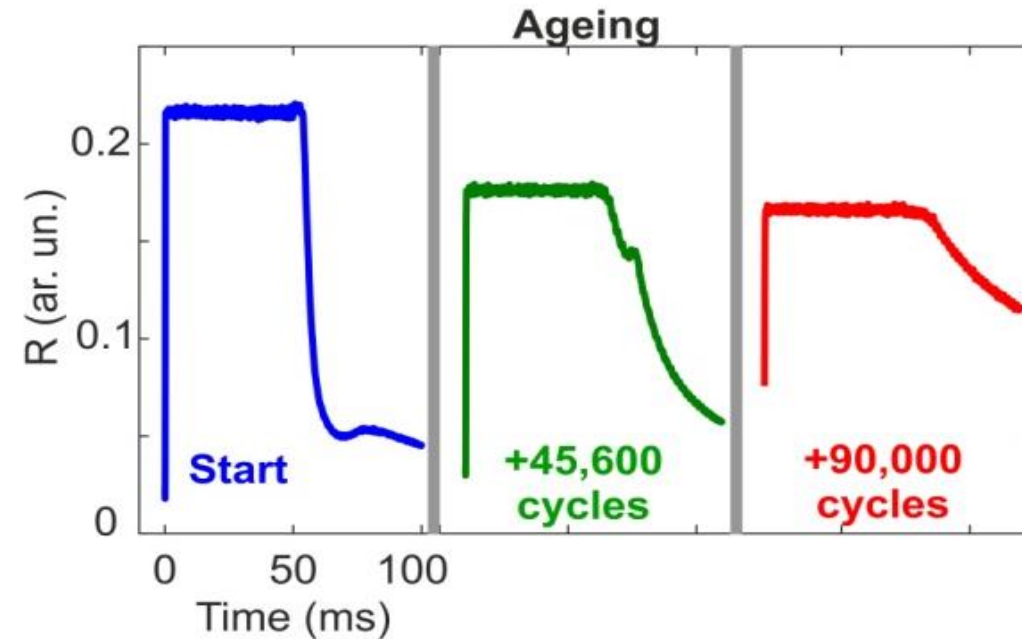
Diffuse reflection



Laser ON → Membrane stretched



Specular reflection



Bartłomiej
Graczykowski,
AMU Poznan

Very fast, durable response

**Potential applications as sensors or
nano-actuators**

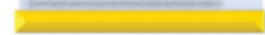
Thin PDA Films for Filtration Applications

Transfer process

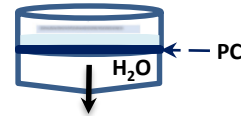
PDA on Au



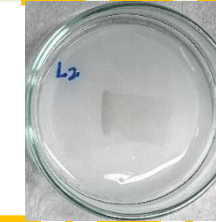
PDA/PVA on Au



PDA/PVA



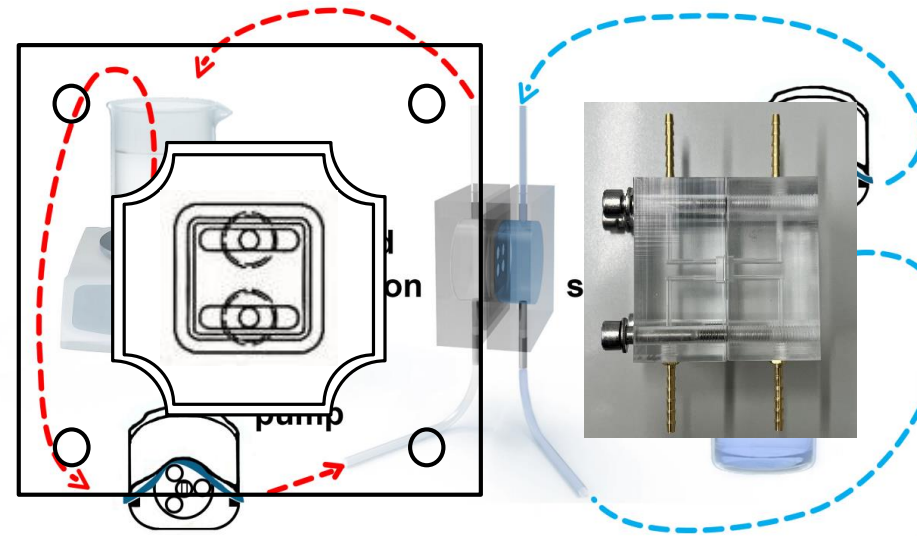
PDA on PC membrane



Pressure driven filtration system



Forward osmosis filtration system



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